

REVIEW

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Navigating AI and machine learning in cancer research: an end-to-end translational framework

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Abstract

Cancer is a complex and heterogeneous disease that is characterized by multi-level biological variability. Advances in high-throughput technologies have led to large-scale, high-dimensional data sets in cancer research, creating a pressing need for powerful computational techniques for successful data analysis. Current techniques may be inadequate for this purpose, thus underscoring the potential of artificial intelligence (AI) and machine learning (ML) for successful data analysis. This review provides a comprehensive pipeline for artificial intelligence/machine learning in cancer research, including preclinical research, clinical decision support, and real-world implementation. It emphasizes several important technologies, data integration, and implementation challenges. The review critically examines multi-omics fusion architectures, regularization-based machine learning, batch-effect harmonization, explainable AI, and federated learning, while addressing translational barriers including algorithmic bias, covariate drift, and regulatory asynchrony across Indian, US, and EU frameworks. Anchored by Decision Curve Analysis as a clinical utility benchmark, this narrative framework establishes that meaningful progress in precision oncology, early detection, and patient outcomes demands not only predictive accuracy but also externally validated, population-representative, and governance-compliant AI systems capable of sustained real-world oncology impact.

Keywords Artificial intelligence, Machine learning, Precision oncology, Cancer research, Clinical decision support, Multi-omics integration, Big data

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Global cancer burden and data

Current global estimates suggest that cancer is among the most prominent contributors to morbidity and mortality, with great variability in incidence and mortality rates across various countries, age, and sex. The synthesis of data from various cancer sites and sources reveals that cancer is not one disease but rather many, with great biological variability driven by studies of demographic, environment, and healthcare factors. The simultaneous inclusion of various cancer types, broad age stratification, and regional variability leads to complex data sets that cannot be analyzed using traditional tools of statistical analysis [1]. Therefore, it is essential to use data-driven



methods of synthesizing trends in cancer incidence and biological data in an attempt to optimize risk assessment and accelerate transition research towards optimizing cancer control [2].

Global burden of cancer: trends, disparities, and future projections

Global cancer epidemiology over the past three decades reveals a divergence between rising case counts and falling age-standardized mortality. In 2021, the number of new cancer incidence and deaths is estimated to be 23.6 million and 9.8 million, respectively. An enormous contribution to the increased number is obtained because of the aging and growing population. Though there is a slight increase in the incidence rate, a very large decline in the age-standardized mortality rate is observed [3]. Future estimates indicate that the incidence and prevalence of cancer will continue to increase worldwide by 2050, particularly among older generations, whereas mortality rates and DALYs will steadily decline [3]. Recent estimates from the World Health Organization and the International Agency for Research on Cancer indicate cancer to be a leading cause of morbidity and mortality. In 2022, it was postulated that 20 million cases of cancer occurred worldwide, with 9.7 million deaths; more than 53 million lives were at risk due to a cancer diagnosis within five years. Despite the progress made in

the sphere of cancer control, access to basic services is uneven. Projections based on this level of progress suggests that by 2050, the worldwide cancer incidence will be more than 35 million new cases annually, an increase from current levels that is far greater than observed in the past. The absolute increase will be largest in high-income countries, while the fastest proportional increases will be among the low and middle HDI countries. These trends highlight the urgent need for equitable investment in cancer prevention, early detection, treatment, and palliative care worldwide. The challenges and lessons associated with the cancer big data pipeline are summarized in Table 1.

Statistical limitations of conventional methods in high-dimensional oncology data

Classical oncology statistics Cox proportional-hazards regression, ordinary least squares, and standard logistic regression are architecturally constrained by the requirement that the number of observations n substantially exceeds the number of predictors p . Modern high-throughput genomic, transcriptomic, and radiomic platforms systematically violate this assumption, generating 10^4 to 10^6 molecular features per patient from cohorts of only 10^2 to 10^3 individuals. This inversion represents a structural incompatibility, not merely a practical inconvenience. For a cohort with D uncensored events at ordered event times $t < \dots < t_D$ and risk set $R(t_i)$ at each event time, the Cox partial likelihood is:

$$\hat{\beta}_{\text{EN}} = \arg \min_{\beta} \left\{ -\ell(\beta) + \lambda \left[\alpha \|\beta\|_1 + \frac{1}{2}(1 - \alpha) \|\beta\|_2^2 \right] \right\}, \quad \alpha \in [0, 1] \quad (1)$$

where:

- $L(\beta)$ is the partial likelihood as a function of the parameter vector β ;
- β is the p -dimensional vector of regression coefficients;
- x_i is the p -dimensional covariate vector of the i -th patient at risk; the superscript T denotes vector transpose;
- t_i is the time of the i -th uncensored event;
- D is the total number of uncensored events in the cohort;
- $R(t_i)$ is the risk set, that is, the set of patients still under observation just prior to event time t_i .

Taking logarithms yields the log partial likelihood,

$$\ell(\beta) = \sum_{i=1}^D \left[x_j^T \beta - \log \sum_{j \in R(t_i)} \exp(x_j^T \beta) \right] \quad (2)$$

Table 1 Major challenges and strategic solutions for cancer big data integration, analytics, and clinical translation

Challenge Area	Insights	References
Data quality and interoperability	Standardized formats, common data models, and consistent collection are essential for reliable and comparable analysis.	[1, 2, 4]
Scalability & Data Aggregation	Cloud-based and federated architectures enable efficient handling of large, continuously updating datasets.	[5–7]
Workflow Integration	Embedding data capture into routine clinical workflows improves adoption and minimizes clinician burden.	[4, 7–9]
Data Completeness	Integrating multi-modal data (clinical, molecular, imaging, outcomes) enhances predictive and translational value.	[4, 6, 7, 10]
Quality Assurance (Qa/Qc)	Continuous and early-stage validation ensures data reliability and reduces downstream analytical errors.	[3, 4, 7, 11]
Data Governance & Sharing	Clear policies on privacy, consent, and access are critical for secure and effective data sharing.	[7, 10, 12, 13]
Analytical Validity	Robust statistical methods and transparent modelling are necessary to avoid bias and false associations.	[3, 4, 11, 12]
Infrastructure & Standards	Interoperable systems, APIs, and modern data standards improve reproducibility and long-term sustainability.	[4, 5, 7, 11, 14]

where $\ell(\beta)$ is the log partial likelihood. All other symbols are as defined for Eq. 1.

Maximum partial-likelihood estimation requires inverting the observed information matrix $I(\beta) = -\nabla^2 \ell(\beta)$, a $p \times p$ matrix. Because each uncensored event contributes at most rank-1 information through its covariate vector x_i , the rank of $I(\beta)$ is bounded by $D \leq n$. Under genomic $p > n$ conditions:

$$\text{rank}(I(\beta)) \leq D \leq n < p \Rightarrow I(\beta) \text{ is singular and not invertible} \quad (3)$$

where $I(\beta)$ is the observed Fisher information matrix equal to the negative Hessian of $\ell(\beta)$; $\text{rank}(\cdot)$ denotes matrix rank; n is the patient count; and p is the predictor count. $I(\beta)$ is therefore singular and non-invertible: the maximum partial-likelihood estimator either fails to exist (complete separation) or exhibits infinite variance, yielding hazard-ratio estimates with no external validity. Ordinary least squares suffer an analogous rank degeneracy. The clinical consequence is not statistical inefficiency but analytical breakdown genome-scale Cox regression without regularization is mathematically ill-posed.

Regularization restores identifiability by appending a convex penalty to the negative log partial likelihood. The LASSO-Cox, ridge-Cox, and elastic-net Cox estimators are:

$$\hat{\beta}_{LASSO} = \arg \min_{\beta} \{-\ell(\beta) + \lambda \|\beta\|_1\}, \quad \|\beta\|_1 = \sum_{j=1}^p |\beta_j| \quad (4)$$

$$\hat{\beta}_{ridge} = \arg \min_{\beta} \{-\ell(\beta) + \lambda \|\beta\|_2^2\}, \quad \|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2 \quad (5)$$

$$\hat{\beta}_{EN} = \arg \min_{\beta} \left\{ \ell(\beta) + \lambda \left[\alpha \|\beta\|_1 + \frac{1}{2} (1 - \alpha) \|\beta\|_2^2 \right] \right\}, \quad \alpha \in [0, 1] \quad (6)$$

where:

- $\hat{\beta}_{LASSO}$, $\hat{\beta}_{ridge}$ and $\hat{\beta}_{EN}$ are the LASSO-Cox, ridge-Cox and elastic-net Cox coefficient estimates;
- $\ell(\beta)$ is the log partial likelihood from Eq. 2;
- $\lambda > 0$ is the regularization strength, selected by k-fold cross-validation;
- $\|\beta\|_1$ is the ℓ_1 norm of β , defined as the sum $\sum_{j=1}^p |\beta_j|$;
- $\|\beta\|_2^2$ is the squared ℓ_2 norm of β , defined as the sum $\sum_{j=1}^p \beta_j^2$;
- $\alpha \in [0, 1]$ is the mixing parameter that balances ℓ_1 sparsity against ℓ_2 stability;
- j indexes the p individual coefficients of β .

LASSO-Cox achieves simultaneous estimation and variable selection by shrinking irrelevant coefficients to

exactly zero biologically equivalent to automatic pathway filtering. Ridge-Cox stabilizes estimation in highly correlated predictor blocks characteristic of co-expressed gene modules. Elastic-net Cox combines both mechanisms and remains the methodological default for correlated multi-omics feature matrices [15, 16].

Beyond linear regularization, non-linear dimensionality reduction and ensemble survival learners' random survival forests, gradient-boosted survival trees, and deep survival networks additionally accommodate non-additive feature interactions, time-varying covariate effects, and proportional-hazards violations endemic to heterogeneous cancer transcriptomes [17]. These mathematical properties, rather than data volume alone, establish advanced computational methods as an analytical necessity in modern translational oncology, where genomic $p \gg n$ is the rule rather than the exception.

AI-driven multi-omics approaches in cancer treatment

Modern oncology has increasingly shifted toward integrative, omics-driven research to better capture the biological complexity of cancer. By combining genomic, transcriptomic, proteomic, metabolomic, and epigenomic information, researchers can examine tumors across multiple regulatory layers, allowing for more nuanced insights into tumor initiation, progression, and heterogeneity. This systems-level perspective has become essential for biomarker discovery and for refining patient stratification strategies. The rapid expansion of high-dimensional omics datasets has driven the adoption of artificial intelligence and machine learning approaches capable of extracting clinically meaningful patterns from complex data. These computational methods have demonstrated value in improving diagnostic accuracy, modelling disease trajectories, and supporting individualized therapeutic decision-making by integrating molecular profiles with clinical and demographic information. In particular, predictive models derived from multi-omics data provide a more detailed assessment of patient-specific risk and treatment response than single-layer analyses. Despite substantial progress, the translation of AI-enabled multi-omics research into routine clinical practice remains constrained by data heterogeneity, limited availability of well-annotated datasets, interpretability of computational models, and regulatory considerations [12].

Multi-omics integration follows three distinct architectures: early fusion concatenates raw features before model training; intermediate fusion projects modality-specific representations into a shared latent space via autoencoders, attention mechanisms, or contrastive learning objectives, currently dominating survival-prediction benchmarks; and late fusion combines

independent unimodal predictions at the decision level, preserving modality-missing robustness at the cost of cross-modal interaction depth. Explainable AI and digital twin models further strengthen interpretability and support personalized precision oncology [18, 19].

Despite these advances, clinical translation remains constrained by batch effects, missing data, high dimensionality, and cross-institutional platform variability. Uncorrected site-specific confounders cause models to learn technical artifacts rather than biologically meaningful oncogenic signatures the primary driver of external-validation collapse. Algorithmic harmonization is therefore mandatory: ComBat applies empirical-Bayes shrinkage to correct additive and multiplicative batch effects, while domain-adversarial neural networks learn site-invariant representations through adversarial min-max optimization. Furthermore, standard AUC-based validation is analytically insufficient for class-imbalanced oncology datasets; precision-recall analysis, F1-scoring, and cost-sensitive learning that explicitly penalizes false negatives are essential for rare mutational subtypes and skewed survival distributions [18, 19].

Modelling approaches in oncology

Modelling approaches in oncology are commonly classified into bottom-up and top-down methods. Although bottom-up approach provides valuable mechanistic insight, their complexity and computational demands limit their direct applicability in routine clinical outcome prediction.

Top-down models typically employ supervised learning when clinical outcomes are known and unsupervised learning to explore data structure, identify patient subgroups, and reduce dimensionality. Logistic regression is frequently applied for binary outcomes due to its simplicity and interpretability, while machine learning methods are used to capture more complex relationships. However, concerns regarding model interpretability have limited clinical adoption, leading to increased interest in probabilistic graphical models such as Bayesian networks, which provide more transparent representation of variable relationships and uncertainty [4].

AI-driven approaches in personalized cancer medicine

Artificial intelligence is increasingly shaping personalized cancer medicine by enabling efficient analysis of large and heterogeneous datasets, including genomic, imaging, and clinical data. AI-based approaches support the identification of therapeutic targets and prediction of treatment response, contributing to more individualized cancer care. Despite growing clinical interest, the implementation of personalized cancer medicine remains challenging due to limited access to targeted therapies, regulatory and financial constraints, and difficulties in

designing individualized clinical trials. Emerging models such as precision medicine tumor boards and patient-centered care frameworks highlight the importance of multidisciplinary collaboration and active patient involvement. Overcoming organizational, technical, and ethical barriers will be critical to fully realize the clinical potential of AI-driven personalized cancer medicine [10]. An integrated artificial intelligence and machine learning (AI/ML) framework for cancer big data analysis is illustrated in Fig. 1.

Data integration architecture: early, intermediate, and late fusion

Multi-omics integration architectures fall into three categories, each with fundamentally distinct computational trade-offs that become most consequential when combining highly sparse molecular data such as binary somatic mutation matrices with high-density spatial data, such as whole-slide histopathology images or volumetric radiological tensors.

Early (data-level) fusion concatenates raw or minimally transformed features from all modalities into a single input vector before model training. While this maximizes theoretical cross-modal interaction modelling, dense radiomic or pathomic tensors inherently overwhelm sparse genomic mutation matrices unless rigorous per-modality variance stabilization and aggressive dimensionality reduction are applied. In practice, naïve early fusion propagates the noise and dimensionality of the largest modality into the unified feature space, systematically suppressing biologically meaningful low-frequency mutational signals [20].

Intermediate (joint-representation) fusion learns modality-specific encoders through autoencoders, multi-omics factor analysis (MOFA, MOFA+), attention-based transformers, or cross-modal contrastive objectives that project each data type into a shared latent space before downstream prediction. This paradigm best preserves the statistical structure of sparse mutation data alongside dense imaging representations, and currently dominates survival- and treatment-response prediction benchmarks [20–22]. Its primary vulnerability lies in encoder sensitivity to batch effects; without explicit harmonization, learned representations risk encoding site-specific technical variation rather than genuine tumour biology.

Late (decision-level) fusion trains independent unimodal models and combines outputs via stacking, weighted averaging, or meta-learners. It is inherently robust to missing modalities a clinically critical property when complete multi-omics profiles are unavailable at the point of care but cannot capture fine-grained cross-modal dependencies that drive tumour heterogeneity. Hybrid and gated-fusion architectures that adaptively reweight modalities at inference are increasingly

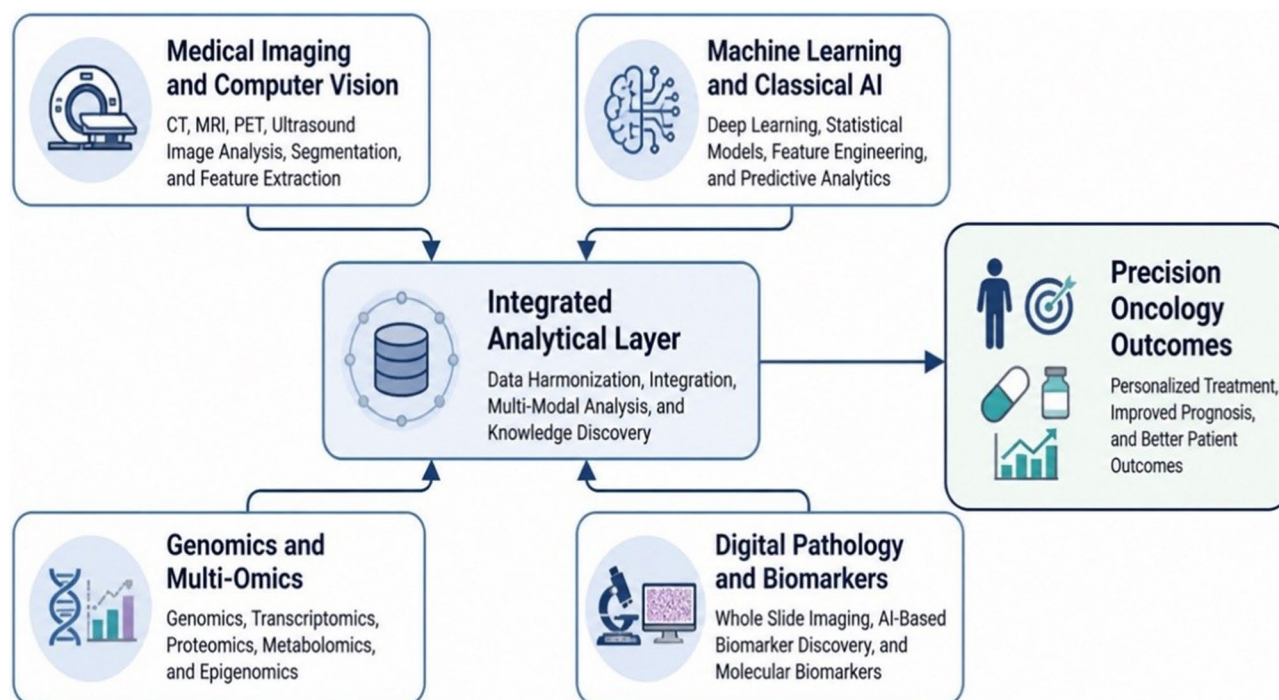


Fig. 1 Integrated AI/ML framework for precision oncology. Multi-modal datasets obtained from medical imaging, genomics and multi-omics, digital pathology, and machine learning platforms are integrated through a centralized analytical layer for data harmonization and knowledge discovery. The framework enables biomarker identification, patient stratification, therapeutic response prediction, and clinical decision support, ultimately advancing personalized cancer management and precision oncology outcomes. Generated using <https://www.canva.com/graphs/>

preferred when modality quality varies systematically across patients or institutions [23].

Necessity of integration of AI/ML

Global and Indian cancer prevalence

Beyond aggregate incidence, the clinical and economic weight of cancer falls unevenly across health systems and populations. The combined effect of demographically altering factors like transition, aging population, urbanization, and risk factors has contributed largely to the constantly increasing global prevalence of cancer [5]. This burden continues its upward trajectory, propelled by an ageing global population, rapid urbanisation, shifting environmental exposures, tobacco consumption, unhealthy dietary patterns, and escalating rates of obesity-related malignancies. Breast, lung, colorectal, prostate, and gastric cancers dominate incidence statistics worldwide, with lung cancer retaining primacy as the leading cause of oncology-related mortality [14]. Despite transformative advances in precision therapeutics and targeted agents, a pronounced divide persists between high-income countries and low- and middle-income countries (LMICs) in early-stage screening coverage, treatment accessibility, long-term survival, and core oncology infrastructure a structural disparity that any data-driven intervention must explicitly account for rather than obscure.

This epidemiological asymmetry disproportionately burdens resource-constrained health systems, where diagnostic backlogs, an overstretched oncology workforce, and restricted access to advanced therapeutic options compound delays across the care continuum [7]. Within India, this challenge is especially acute: diagnostic incidence is projected to escalate sharply over the coming decades, while outcomes remain constrained by late-stage presentations, fragmented referral pathways to tertiary centres, and persistent infrastructural deficits [6, 13]. These structural pressures have driven investment in AI/ML-based triage algorithms, diagnostic decision-support systems, and treatment-stratification pipelines though deployment within heterogeneous, digitally uneven health ecosystems simultaneously introduces documented risks of algorithmic bias, data fragmentation, and unvalidated external generalisability that demand proactive mitigation [13].

Limitations of conventional oncology workflows

The traditional oncology care process is primarily linear, manual, and fragmented in terms of preclinical research, clinical trials, and practice. The data created at various ends of this spectrum, starting from laboratory experiments, animal models, imaging, genomics, and Electronic Health Records, are often fragmented, heterogeneous, and not used to their fullest potential [8].

Additionally, traditional analytical techniques are heavily reliant on human interpretation and hypothesis-driven analysis, which have difficulties in handling the growing complexity of cancer datasets [24]. The high drop-out rates of drugs in clinical pipelines, the length of the clinical trials process, and the lack of personalization of therapies are indicative of systemic inefficiencies in the traditional pipelines [25]. The above points to the inefficacy of traditional pipelines in addressing the challenges of cancer heterogeneity and the potential of integrative computational models [26].

The role of AI/ML in precision and translational oncology

Artificial intelligence and machine learning have fundamentally transformed precision and translational oncology by enabling large-scale integration of heterogeneous clinical, molecular, radiological, and pathological datasets that conventional analytical frameworks cannot practically accommodate [27]. Advanced computational architectures including deep neural networks, graph-based learning systems, and sequence-aware transformer models identify complex nonlinear associations and latent biological patterns undetectable through standard statistical approaches, supporting patient stratification, biomarker discovery, outcome prediction, and treatment-response forecasting [9, 28]. In translational research, AI/ML accelerates target identification, in silico toxicity screening, and preclinical-to-clinical alignment, substantially reducing the attrition that has historically characterised drug development pipelines [29, 30]. However, many high-performing retrospective models demonstrate markedly reduced generalisability during prospective clinical implementation, attributable to hidden confounding variables, demographic imbalance, and dataset shift reinforcing that predictive accuracy alone cannot substitute for rigorous, population-representative external validation [29, 30].

Expanding big data landscape in cancer informatics

Beyond conventional multi-omics integration, emerging big data sources such as polygenic risk score (PRS) models, liquid biopsy datasets, and multimodal clinical records are expanding the scope of AI-driven oncology research. PRS approaches utilize large-scale genomic variant data to estimate an individual's inherited susceptibility to cancer, enabling population-level risk stratification and preventive strategies. In parallel, AI analysis of liquid biopsy data, including circulating tumor DNA and protein biomarkers, offers a minimally invasive method for early detection and continuous disease monitoring. Recent advances in multimodal learning further enable integration of heterogeneous data types such as radiological images and unstructured clinical text, improving the depth and reliability of predictive modelling.

Additionally, large population-scale screening programs are generating real-world datasets that enhance model robustness and support translation of big data analytics into practical cancer care settings, particularly in resource-constrained healthcare systems [31].

Heterogeneous ecosystems and socio-infrastructure variations

The principal barrier to clinical AI adoption in developing healthcare landscapes such as India is not a deficit in algorithmic sophistication, but systematic heterogeneity across regional infrastructure and digital health maturity. A model demonstrating strong internal validity on standardised, digitally integrated tertiary-centre data exhibits marked performance degradation when exposed to non-standardised registries, inconsistent imaging formats, variable staining protocols, and fragmented health records characteristic of rural and semi-urban facilities [32]. These disparities generate covariate shifts that trained models cannot absorb without explicit domain adaptation, federated learning, or transfer-learning strategies computational safeguards that must be architecturally embedded from the outset rather than applied as post-deployment corrections [32].

Motivation: overcoming research-implementation gaps

Despite substantial research progress, a persistent and well-documented gap separates algorithm development from clinical deployment in oncology AI. Models demonstrating strong retrospective performance consistently underperform during prospective implementation a pattern attributable to overfitting on curated single-institution datasets, unrecognised confounding variables, and distribution shift between training and deployment populations [33, 34]. The failure of IBM Watson for Oncology to achieve consistent treatment concordance across diverse health systems exemplifies how algorithms optimised on academically controlled corpora collapse when confronted with real-world clinical heterogeneity [33]. Absence of external validation, insufficient model interpretability, and clinician acceptance barriers represent the primary mechanistic causes of this translational failure [34]. Within India, these challenges are compounded by infrastructure diversity, inconsistent data quality, and variable digital maturity across facilities [35]. Bridging this divide requires a coordinated pipeline encompassing prospective validation, independent algorithmic auditing, post-deployment drift monitoring, and robust governance frameworks embedding AI as continuously maintained clinical infrastructure rather than isolated point-of-need solutions [36, 37].

Research gap and scope of the present review

Artificial intelligence and machine learning have achieved rapid penetration across cancer research domains including medical imaging, biomarker discovery, genomics, radiomics, and multimodal data analysis. Existing narrative reviews have documented the application of convolutional neural networks, transformers, graph neural networks, and generative models for cancer diagnosis and prognosis. However, the predominant focus of these reviews remains on individual technical applications or algorithm benchmarking, with limited systematic examination of how these technologies can be integrated across the complete translational oncology continuum from preclinical investigation through regulatory approval and post-deployment monitoring in operationally realistic clinical environments [38, 39]. Furthermore, comparative regulatory analysis spanning the Indian CDSCO framework, the FDA Software as a Medical Device guideline, and the EU AI Act remains largely absent from existing literature, leaving implementation-oriented practitioners without actionable cross-jurisdictional guidance.

This narrative review addresses these gaps by providing a structured end-to-end AI/ML framework for translational oncology that deliberately integrates technical, clinical, regulatory, and implementation perspectives within a single coherent workflow. Distinct from prior algorithmic surveys, its unique contribution lies in explicitly connecting multimodal data harmonisation, model interpretability, clinical utility assessment, and adaptive governance with particular emphasis on implementation challenges and opportunities within the Indian healthcare landscape into a practically deployable translational architecture [39].

End-to-end AI/ML solution pipeline for cancer treatment

Data foundation layer

Receipt of data

Data acquisition is the starting point of the workflow, and it refers to the organized and systematic retrieval of disparate data with respect to cancer biology studies conducted in the preclinical and clinical setting [36] and represented by a variety of data types including multi-omics platforms (genomic, transcriptomic, proteomic data) and digital images of radiology and digital pathology, clinical trial data, electronic records, and population-level registers [8]. Large-scale efforts like TCGA, ICGC, and Genome India, and the National Cancer Grid, among others, have leveraged data and allowed the creation of cohorts amenable to analysis by artificial intelligence including machine learning algorithms and analytics techniques based on machine learning and deep learning approaches, machine vision [25, 35]. A schematic representation of the end-to-end artificial intelligence

and machine learning (AI/ML) pipeline, illustrating key stages from data acquisition to real-world deployment, is depicted in Fig. 2.

Data ingestion, governance, and storage

After acquisition, the data has to be ingested and governed securely for integrity, traceability, and regulation compliance [37]. Data governance frameworks are based on FAIR principles and rely on standardized clinical and image data schema for future interoperability [6]. Storage systems with scalable infrastructures, including cloud computing or high-performance computing environments, are needed for dealing with petabyte-sized genomics and image databases [29].

Data preparation layer

Data preprocessing and harmonization

Real-world oncology datasets are very often noisy, incomplete, and dataset- and institution-specific. Normalization, removal of personally identifiable information, and artifact correction are part of preprocessing pipelines in both single and multimodal datasets [36]. Normalization, adjustment for differences in imaging equipment, alignment, variant detection, and removal of batch effects are carried out in imaging and genomics datasets respectively and are standard procedures in these domains [24].

Multi-centric oncology datasets are routinely confounded by site-, scanner-, platform-, and protocol-specific variation collectively termed batch effects or domain shift that can systematically overwhelm the underlying biological signal of interest. Left uncorrected, these technical confounders cause predictive models to encode institutional signatures rather than tumour biology, representing the primary documented cause of external-validation collapse in oncology AI. Three algorithmically distinct families address this challenge. First, empirical-Bayes methods such as ComBat model the observed value $Y_{ij,g}$ for feature g in sample j of batch i as

$$Y_{ij,g} = \alpha_g + X\beta_g + \gamma_{i,g} + \delta_{i,g} \cdot \varepsilon_{ij,g} \quad (7)$$

where:

- $Y_{ij,g}$ is the measured value of feature g in sample j of batch i ;
- α_g is the overall mean of feature g across all samples;
- X is the design matrix encoding the biological covariates that must be preserved;
- β_g is the effect vector associated with X for feature g ;
- $\gamma_{i,g}$ and $\delta_{i,g}$ are the additive and multiplicative batch effects for feature g in batch i , estimated by empirical-Bayes shrinkage toward feature-set hyperparameters [40].
- $\varepsilon_{ij,g}$ is a zero-mean error term.

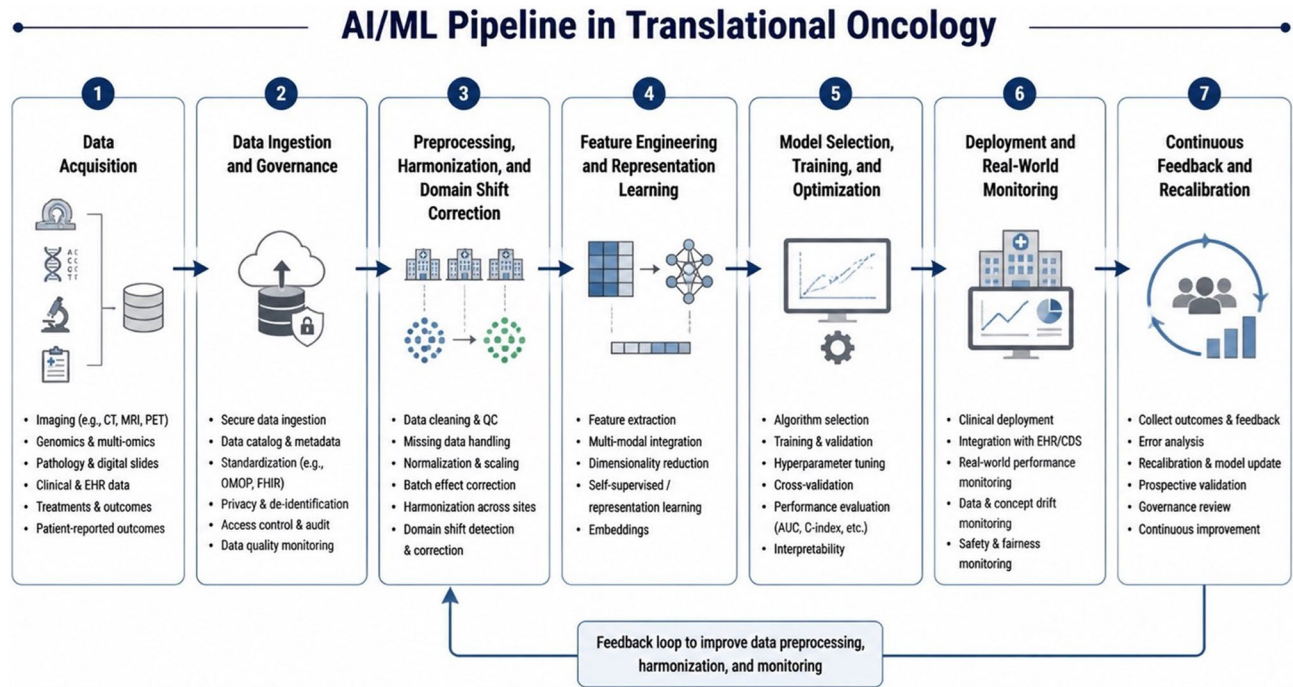


Fig. 2 End-to-end artificial intelligence and machine learning (AI/ML) pipeline for translational oncology. The workflow demonstrates successive stages of AI/ML implementation, encompassing multi-modal data acquisition, secure data integration and governance, preprocessing and harmonization, feature extraction and representation learning, model development and optimization, clinical deployment, and real-world performance monitoring. Continuous feedback and recalibration enable iterative model refinement, ensuring robustness, interpretability, and clinical applicability for precision oncology, biomarker discovery, therapeutic decision-making, and outcome prediction. Generated using <https://www.canva.com/graphs/>

ComBat-seq extends this formulation to negative-binomial-distributed RNA-seq counts, explicitly accommodating the overdispersion and library-size heterogeneity intrinsic to high-throughput count data [41]. Second, latent-variable methods SVA, RUV, PEER, and Harmony infer and remove unobserved technical confounders directly from data structure without requiring known batch labels, rendering them applicable when institutional metadata is incomplete or entirely unavailable. Third, deep domain-adaptation methods Domain-Adversarial Neural Networks (DANN), Deep CORAL for second-order feature-distribution alignment, and cycle-consistent generative networks for cross-scanner MRI and cross-stain histopathology harmonisation cast the correction as a minimax optimisation between a feature extractor f_θ , a task classifier h_φ , and a domain discriminator g_ψ :

$$\min_{\theta, \varphi} \max_{\psi} [L_{task}(h_\varphi \circ f_\theta) - \lambda \cdot L_{domain}(g_\psi \circ f_\theta)] \quad (8)$$

where:

- f_θ is the feature extractor parameterised by θ ;
- h_φ is the task classifier parameterised by φ , predicting the clinical outcome;
- g_ψ is the domain discriminator parameterised by ψ , predicting site or batch identity;

- L_{task} is the supervised task loss (for example, cross-entropy on the clinical label);
- L_{domain} is the domain-classification loss;
- $\lambda > 0$ controls the trade-off between task accuracy and site invariance;
- $\circ$ denotes function composition.

Robust harmonisation prior to downstream modelling is not optional but analytically mandatory uncorrected batch effects represent the most consistently documented cause of external-validation collapse in multi-centric oncology AI. For structured high-dimensional molecular data, including gene expression profiles, DNA methylation arrays, and radiomic feature matrices, empirical-Bayes methods such as ComBat and ComBat-seq remove additive and multiplicative site effects while explicitly preserving covariate-encoded biological signal through known-variable modelling. When batch labels are unavailable, latent-variable approaches SVA, RUV, and Harmony estimate and regress unobserved technical factors directly from data structure. For non-tabular spatial modalities, including radiological imaging and whole-slide digital pathology, adversarial domain-adaptation frameworks such as DANN and Deep CORAL generate site-invariant feature representations, with the learned encoder remaining statistically indistinguishable across institutional domains while retaining clinical

predictive signal [42]. Regardless of method selected, routine reporting of batch-effect diagnostics principal-variance component analysis, gPCA, and kBET alongside model performance metrics constitutes a minimum transparency standard for multi-centric oncology AI.

Feature engineering and representation learning

This involves the processing of data into useful models, and there are two ways in which this is achieved: by feature engineering, where features such as radiomic textures and gene expression signatures are extracted, while deep learning focuses on end-to-end representation learning from input data [28]. Representation learning plays an important role especially in multimodal fusion. It enables the embedding of diverse data types into a common space where meaningful relationships across domains can be maintained [9]. Representation learning is seen as one of the basic building blocks for predictive modelling in either pre-clinical as well as clinical environments [5].

Model development

Selection of AI/ML algorithm

The choice of algorithm depends on data modality, level of complexity, and medical intent. CNNs dominate imaging, Transformers are used for sequence and text data, and GNNs model for chemical interactions and the graphical structures of life, as reported [7, 13, 14]. Generation networks have their uses in de novo synthesis or data augmentation, while conventional machine learning for structured table data has its applications [28]. The choice process needs a balance between predictive accuracy and interpretability, computability, and legal requirements [26].

Model training and optimization

For the training of the model, there is the use of iterative optimization through the use of labelled datasets, and there may be the use of transfer learning or self-supervised pretraining to overcome data constraints [33]. Hyperparameter tuning, imbalance correction, and regularization techniques are used to boost the ability to be more generalizable [25]. Privacy-preserving models, such as Federated Learning, can be used to train models collaboratively across multiple sites without needing to share patient data, thereby overcoming the limitations of regulations and ethics [30].

Evaluation and insight layer

Validation and performance metrics

Sensitivity, specificity, and AUC-ROC, although universally reported, are fundamentally unreliable in class-imbalanced oncology tasks such as identifying rare driver mutations, early-stage malignancies, or

treatment-responder subgroups where positive-class prevalence may fall below 1–5%. AUC-ROC is mathematically insensitive to class prevalence: at a 1:99 imbalance ratio, a naïve classifier predicting all negatives achieves 99% accuracy yet zero clinical utility. Precision-Recall AUC, the F1-score, and the Matthews Correlation Coefficient (MCC) are therefore essential primary reporting endpoints. In high-stakes diagnostic contexts where a missed malignancy carries irreversible clinical consequences, cost-sensitive learning frameworks that explicitly assign asymmetric misclassification penalties heavily penalising false negatives relative to false positives are analytically necessary rather than optional enhancements; rigorous selection of appropriate performance parameters and validation across diverse structural datasets are paramount for ensuring generalisability of novel computational frameworks [43]. With precision $Pr = TP / (TP + FP)$ and recall $Re = TP / (TP + FN)$, the F1-score is the harmonic mean of Pr and Re ,

$$F1 = 2 \cdot \frac{Pr \cdot Re}{Pr + Re} \quad (9)$$

where:

- F1 is the F1-score, a harmonic mean of precision and recall;
- Pr is precision, defined as $Pr = TP / (TP + FP)$;
- Re is recall (also called sensitivity), defined as $Re = TP / (TP + FN)$;
- TP , FP and FN are the counts of true positives, false positives and false negatives in the confusion matrix.

MCC, which incorporates all four confusion-matrix quadrants in a single balanced score, is defined as

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (10)$$

where:

- MCC is the Matthews Correlation Coefficient, bounded in $[-1, +1]$ with 0 indicating no better than random;
- TP , TN , FP and FN are the counts of true positives, true negatives, false positives and false negatives, respectively.

MCC is the most robust single-number metric for imbalanced binary classification because all four cells of the confusion matrix enter its computation [44, 45]. Cost-sensitive learning strategies (class-weighted cross-entropy, asymmetric misclassification cost matrices, and the focal loss) are deployed whenever false negatives

carry disproportionate clinical harm. With p_t the model's predicted probability for the true class and $\gamma \geq 0$ a focusing parameter, the focal loss is

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \cdot \log(p_t) \quad (11)$$

where:

- $FL(p_t)$ is the focal-loss value for a single training example;
- p_t is the model's estimated probability assigned to the true class for that example;
- $\alpha_t \in (0, 1]$ is a class-balancing weight, typically set inversely proportional to class frequency;
- $\gamma \geq 0$ is the focusing parameter; $\gamma = 0$ recovers standard cross-entropy, and $\gamma = 2$ is the value used by Lin et al. in the original report;
- the factor $(1 - p_t)^\gamma$ down-weights easy, well-classified majority examples and concentrates gradient updates on hard, rare-class examples [46].

Beyond discrimination and class-imbalance metrics, models intended for clinical deployment must additionally report calibration quantified through the Brier score, Expected Calibration Error (ECE), and reliability diagrams alongside net clinical benefit through Decision Curve Analysis. The principle that rigorous selection of descriptive parameters and comprehensive validation across structurally diverse datasets are prerequisites for predictive accuracy and generalisability, demonstrated in computational chemistry and adjacent molecular modelling domains [47], applies with equal force to translational oncology frameworks.

Despite widespread reporting of sensitivity, specificity, and AUC-ROC, these discrimination metrics do not account for the real-world consequences of acting on model outputs. A model with a high AUC can still recommend unnecessary interventions or fail to alter decisions that meaningfully affect outcomes. This structural disconnect the performance-utility gap underscores that statistical optimality and clinical usefulness are distinct, independently measurable properties, and that evaluation frameworks must explicitly quantify both to avoid deploying statistically impressive but clinically inert systems [48].

Decision Curve Analysis (DCA) addresses this gap by quantifying net benefit across a spectrum of clinically plausible threshold probabilities. Net benefit is formally defined as:

$$NB = TPR - FPR \times \left(\frac{p_t}{1 - p_t} \right) \quad (12)$$

where p_t is the threshold probability at which clinical intervention is judged appropriate. This formulation simultaneously penalises false positives unnecessary interventions and rewards true positive identifications, embedding the clinician's implicit harm-benefit trade-off directly into the metric rather than treating it as an external consideration [48].

In a standard decision curve, the x-axis represents the threshold probability and the y-axis represents net benefit. Two reference strategies define the comparative baseline: treating all patients unconditionally and treating none. A model demonstrates genuine clinical value only when its decision curve consistently exceeds both baselines across the clinically relevant threshold range. Models exhibiting high discrimination yet flat or negative net benefit curves lying below either reference line must be regarded as clinically non-viable regardless of their AUC or F1-score [49].

In oncology, treatment decisions require explicitly balancing the harms of overtreatment against the irreversible consequences of missed diagnoses precisely the trade-off DCA formalises. A model with exceptional AUC may deliver zero net benefit if its threshold recommendations generate over-intervention in low-risk patients or under-treatment in high-risk subgroups. DCA therefore functions as the definitive translational filter separating statistically successful models from clinically deployable decision-support tools [48, 49].

Model interpretability and explainability

Interpretability mechanisms provide insight into how models make decisions. This helps build trust among clinicians, which help to obtain regulatory approval [26]. Techniques like attention mapping, saliency analysis, and feature attribution identify the data elements that influence predictions [35]. Explainability is particularly important in high-stakes oncology applications. Opaque models may prevent adoption, even if they perform well [27].

Deployment and influence layer

Clinical integration and decision support

Deployment enables the integration of models that have been validated into clinical information systems such as electronic records and picture archiving and communication systems (PACS), as well as clinical decision support systems [37]. Seamless integration into the workflow reduces the clinicians' cognitive workload and supports the provision of informed insights at the point of care through integration of validated models [28]. In translational settings, decision-support systems have become more prevalent in screening, triage, treatment decision-making, as well as trial-matching [34].

Regulatory considerations for clinical implementation of AI

Regulatory AI for oncology is treated as software that acts like a medical device. It must demonstrate its safety, effectiveness, and strength [38]. These ethical principles include the importance of explainability, overcoming bias, and the need for “human-in-the-loop” oversight and also important for Digital twin concept [50, 51]. The regulatory procedures are not standard across all countries, which makes it necessary to be flexible in validation and documentation processes [35]. Artificial intelligence tools intended for clinical use are generally regulated as Software as a Medical Device (SaMD), requiring demonstration of safety, accuracy, and clinical reliability prior to implementation in healthcare settings. Regulatory agencies such as the U.S. Food and Drug Administration (FDA) evaluate AI systems based on risk classification and approve them through established pathways including 510(k), De novo, or Premarket Approval depending on the intended medical application. As AI models often evolve with new data, regulatory frameworks emphasize a Total Product Life Cycle approach that includes continuous performance monitoring after deployment. Additional considerations such as use of high-quality datasets, transparency of algorithm functioning, and controlled update mechanisms are essential to ensure trustworthy and ethically responsible integration of AI into clinical oncology practice.[52]

Deployment and real-world monitoring deployment monitoring

Post-deployment evaluation and transparent reporting of model performance are very important. It ensures reliability, detect potential biases, and support safe clinical use of AI systems [53]. AI-driven personalized medicine approaches leverage continuously evolving clinical and genomic data to refine patient stratification and improve treatment decision-making over time [43]. This, the final stage, completes the translation loop by using insights derived from deployment to inform the earlier stages of the pipeline [37].

Post-deployment monitoring is a critical yet underdeveloped translational oncology component. Real-world environments introduce continuous performance degradation through two distinct phenomena: covariate shift $P_{\text{deploy}}(X) \neq P_{\text{train}}(X)$ driven by demographic transitions, hardware upgrades, or new sequencing platforms; and concept drift $P_{\text{deploy}}(YX) \neq P_{\text{train}}(YX)$ triggered by revised oncology guidelines, evolving epidemiology, or novel therapeutic protocols [54, 55]. Concept drift is especially insidious as standard input monitors fail to detect it; degradation manifests only through clinical outcome divergence. Both require continuous surveillance via Population Stability Index tracking, Page-Hinkley testing, and automated recalibration supported

by human-in-the-loop oversight. Deployment must therefore be governed as dynamic clinical infrastructure under FDA Predetermined Change Control Plans and EU AI Act adaptive provisions [55].

The operational loop demands continuous adaptive surveillance pipelines tracking Population Stability Indices, prediction-distribution deviations, and calibration trend metrics. Real-time logging infrastructures with automated statistical alerting including sequential Page-Hinkley testing enable frameworks to flag calibration degradation and initiate localised recalibration or scheduled retraining before performance breaches clinically safe thresholds. Embedding structured human-in-the-loop oversight interfaces ensures clinician feedback is systematically captured and operationalised. This collaborative architecture provides a critical patient-safety layer, detecting unexpected bias propagation, edge-case misclassifications, and workflow friction points before they propagate into adverse clinical outcomes [54, 55].

Critical discussion: external validation, bias, drift and real-world impact

External validation failure and interpretability barriers to clinical adoption

Although AI/ML models in oncology report promising retrospective performance, generalisation and reproducibility deficits persistently obstruct clinical translation. External validation studies consistently reveal that institution-specific training datasets fail to represent the phenotypic, demographic, and protocol variability of real-world clinical settings [29, 38]. Models with strong internal performance routinely degrade during prospective evaluation due to domain shift across imaging protocols, patient demographics, and institutional care practices. Inadequate prospective validation remains the most prevalent methodological deficiency in published oncology AI literature [56].

Beyond validation, the limited interpretability of complex deep learning architectures constitutes a fundamental adoption barrier. In high-stakes oncology settings, black-box predictions unsupported by feature attribution methods SHAP, integrated gradients, or saliency mapping linked to biological pathways systematically erode clinician confidence and constrain implementation regardless of predictive accuracy [29, 38]. Explainability, transparency, and workflow compatibility are therefore non-negotiable prerequisites for safe and effective clinical integration of oncology AI systems.

Algorithmic bias propagation and demographic underrepresentation

Dataset bias systematically distorts oncology AI when training repositories overrepresent specific demographic or ancestral groups as in Western-dominated genomic

biobanks that underrepresent South Asian, African, and Hispanic populations. This distributional imbalance propagates into clinically consequential performance disparities across ethnic, socioeconomic, and geographic subgroups, reducing external validity and compromising equitable care delivery [57, 58]. Mitigation requires shifting from standard empirical risk minimisation toward fairness-aware loss functions with mandatory subgroup-stratified external validation reporting.

Characterizing model drift via covariate and concept shift

The real-world deployment of oncology algorithms is inherently susceptible to performance degradation over time, a phenomenon mathematically categorized as model drift. This degradation is driven by two distinct statistical mechanisms: covariate shift, where the input data distribution changes over time,

$$P_{\text{deploy}}(X) \neq P_{\text{train}}(X) \quad (13)$$

and concept shift, where the true conditional probability of the clinical diagnosis alters relative to the feature space,

$$P_{\text{deploy}}(Y | X) \neq P_{\text{train}}(Y | X) \quad (14)$$

In longitudinal clinical environments, these shifts are triggered by evolving diagnostic criteria, the introduction of higher-resolution imaging sensors, shifts in regional referral patterns, or the introduction of novel neoadjuvant systemic therapies. Because deep neural networks are fundamentally static mapping functions, they cannot dynamically adapt to these shifting distributions, leading to progressive calibration decay and inflation of false-negative rates. Consequently, avoiding catastrophic failure modes requires implementing continuous statistical drift-monitoring architectures such as online Page-Hinkley testing or sequential Kullback-Leibler divergence tracking coupled with rigorous automated retraining triggers and real-time recalibration protocols [59].

Prospective clinical evaluation

Despite promising results reported in retrospective studies, prospective clinical evaluation remains essential for determining the real-world effectiveness of AI systems in healthcare. The GEMINI study prospectively evaluated AI integration in routine breast cancer screening across multiple clinical settings and demonstrated that AI-assisted screening could improve cancer detection while reducing radiologist workload. Importantly, the study highlighted that AI performance is influenced not only by the algorithm itself, but also by factors related to clinical implementation, including human oversight, arbitration strategies, and workflow integration. The authors further

emphasized that prospective evaluation provides a more realistic assessment of AI performance under routine clinical conditions by accounting for variations in clinician interaction, operational practices, and institutional settings [60].

Asynchrony in adaptive regulatory frameworks

The clinical translation of oncology software is heavily bottlenecked by an asynchrony between static regulatory oversight mechanisms and the dynamic, iterative nature of machine learning algorithms. Traditional medical device pathways are structurally designed for locked software configurations, categorizing any post-market optimization or online update as a material modification that requires a complete, resource-intensive regulatory re-submission.

While advanced regulatory bodies have begun piloting adaptive strategies such as the United States Food and Drug Administration's (FDA) final guidance on Pre-determined Change Control Plans (PCCP), which permits pre-authorized algorithmic modifications via strict Algorithm Change Protocols global frameworks remain deeply fragmented. Current regulatory architecture is largely unequipped to handle localized continuous learning paradigms, cross-border data privacy mandates (e.g., GDPR compliance within federated networks), and post-market safety tracking for real-time model updates. Bridging this translational gap requires establishing internationally harmonized, total product life cycle (TPLC) regulatory frameworks that treat artificial intelligence as an evolving, risk-stratified software infrastructure rather than a static diagnostic tool. [61–63]

AI/ML technologies enabling the pipeline

Deep learning for imaging (CNNs)

Convolutional Neural Networks (CNNs) provide the paradigm for AI-assisted analysis in radiology and digital pathology in oncology applications [28]. Their ability to extract hierarchical features helps to automatically identify, segment, and classify cancers in imaging modalities such as CT scans, MRI, PET scans, and whole slide images in digital pathology [33]. In translational pipelines, CNN-based radiomics and pathomics identify volumetric features related to tumor grade, genetic changes, and predictive outcomes, facilitating non-invasive biomarkers and decision-making in cancer care [27]. The universality and efficacy of CNNs have led to their pivotal role in studies and initial implementations, especially in screening and diagnostic aid [34].

An overview of modern artificial intelligence and machine learning (AI/ML) technologies applied in drug discovery and biomedical research is illustrated in Fig. 3.

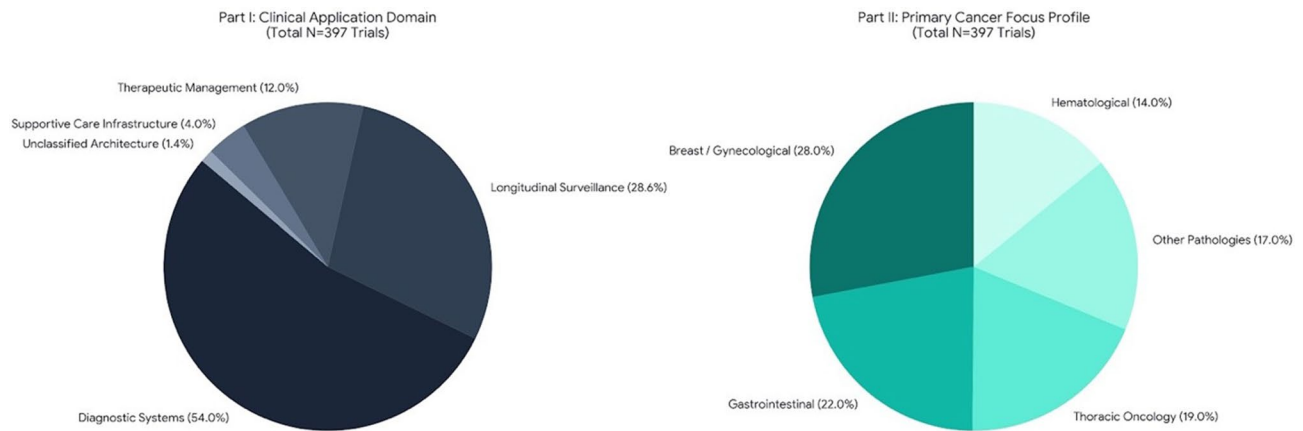


Fig. 3 Clinical distribution of AI/ML applications in oncology trials. Pie charts illustrating the distribution of AI/ML-based oncology clinical trials ($N = 397$) according to application domains and primary cancer focus areas. Diagnostic systems and longitudinal surveillance constitute the largest application sectors, while breast/gynecological, gastrointestinal, and thoracic cancers represent the most frequently investigated disease categories, highlighting the expanding role of AI/ML in precision oncology. Generated using <https://www.canva.com/graphs/>

Transformers and large language models for sequences and text

Transformer models represent an extension of the capabilities of AI algorithms from images to sequential and text data in the field of oncology, including genetic information and trial documentation [14]. Large language models help in extraction, synthesis, and reasoning over complex clinical and biomedical text, which improved utilization of unstructured data in oncology work-flows [24]. Large language models also make it feasible to automatically extract structured data from unstructured clinical notes, thus making EHRs more integrable into real-world evidence [28]. Large language models in translational research help in synthesis, identification of cohort, and alignment of molecular data to the findings in clinical phenotypes [27].

Graph neural networks in molecular and Network biology

Although deep convolutional, graph, and transformer architectures extract predictive features from imaging and multi-omics data with high accuracy, their internal representations remain opaque systematically eroding clinician trust and obstructing regulatory acceptance in high-stakes oncology settings. An uninterpretable prediction, regardless of its discrimination performance, is clinically inactionable and regulatory-audit-resistant. Biologically informed neural architectures address this by constraining network topology to mirror curated biological hierarchies derived from Reactome, KEGG, or Gene Ontology. For a hidden layer l with weights $W^{(l)}$ and a binary biological-topology mask $M^{(l)}$ of dimension $dl + 1 \times dl$, the forward pass is

$$h^{(l+1)} = \sigma \left(\left(M^{(l)} \odot W^{(l)} \right) h^{(l)} + b^{(l)} \right) \quad (15)$$

where:

- $h^{(l+1)}$ is the activation vector at layer $l + 1$;
- $h^{(l)}$ is the activation vector at layer l ;
- $W^{(l)}$ is the learnable weight matrix at layer l ;
- $M^{(l)}$ is a binary mask of the same shape as $W^{(l)}$, whose entries equal 1 where a biological connection exists in the curated ontology and 0 otherwise;
- $\odot$ is the element-wise (Hadamard) product, which zeros out all biologically unsupported connections in $W^{(l)}$;
- $b^{(l)}$ is the bias vector at layer l ;
- σ is a non-linear activation function (for example ReLU or hyperbolic tangent).

Because the mask enforces only biologically supported connections, hidden units in architectures such as P-NET correspond directly to specific Reactome pathways their activations readable as pathway-level evidence for the clinical phenotype, providing mechanistic accountability that is structurally absent from standard deep learning frameworks [64]. Group-LASSO regularisation provides a complementary mechanism for pathway-level sparsification, operating on entire gene sets simultaneously rather than individual coefficients. For G predefined gene sets $\{G_1, \dots, G_G\}$ with corresponding coefficient sub-vectors $\beta = (\beta_{G_1}, \dots, \beta_{G_G})$, the penalty is

$$R(\beta) = \lambda \sum_{g=1}^G \sqrt{|G_g|} \cdot \|\beta_{G_g}\|_2 \quad (16)$$

where:

- $R(\beta)$ is the group-LASSO penalty added to the loss function;

- G is the total number of predefined gene sets (pathways);
- G_g is the g -th gene set, for $g = 1, \dots, G$;
- $|G_g|$ denotes the cardinality of G_g , that is, the number of genes it contains;
- β is the full vector of regression coefficients, partitioned by gene set;
- β_{G_g} is the sub-vector of coefficients corresponding to the genes in G_g ;
- β_{G_g2} is the ℓ_2 norm of β_{G_g} ;
- $\sqrt{|G_g|}$ is a scaling factor that balances unequal gene-set sizes, so that larger sets are not penalised disproportionately;
- $\lambda > 0$ is the regularization strength, selected by cross-validation.

Group-LASSO shrinks entire gene sets simultaneously, producing pathway-level rather than gene-level feature selection that is directly interpretable by oncologists. Comparable pathway-resolved attribution approaches have demonstrated utility in proteomics-based biomarker discovery [65]. Post-hoc attribution methods DeepLIFT, integrated gradients, and SHAP projected onto Reactome and KEGG ontologies further bridge abstract latent representations to mechanistically meaningful oncogenic biology. Implementing robust mathematical descriptors within advanced multi-layered computational frameworks analogously enhances predictive accuracy and convergence stability across molecular modelling domains a principle with established precedent in computational chemistry and structural bioinformatics underscoring the broader value of principled feature representations in hierarchical oncology AI architectures [53, 66].

Generative models for synthetic oncology data: biological fidelity, fairness, and privacy validation

Generative models risk hallucinating biologically non-existent structures and amplifying demographic biases embedded in training cohorts' failure modes with direct consequences for downstream clinical utility. Synthetic multi-omics and imaging data must therefore be validated against a stringent multi-criterion framework before clinical or research deployment. Distributional fidelity between real distribution P and synthetic distribution Q is assessed through three complementary measures; the squared Maximum Mean Discrepancy in a reproducing-kernel Hilbert space with kernel k is:

$$MMD^2(P, Q) = E_{x, x' \sim P} k(x, x') + E_{y, y' \sim Q} k(y, y') - 2E_{x \sim P, y \sim Q} k(x, y) \quad (17)$$

where:

- $MMD^2(P, Q)$ is the squared Maximum Mean Discrepancy between distributions P and Q ;
- P is the real data distribution and Q is the synthetic data distribution;
- $k(\cdot, \cdot)$ is a positive-definite kernel, typically Gaussian or rational quadratic;
- E denotes mathematical expectation under the distribution shown in its subscript;
- x, x' are independent samples drawn from P ; y, y' are independent samples drawn from Q ;
- for characteristic kernels (for example the Gaussian kernel), $MMD^2(P, Q) = 0$ if and only if $p = Q$ [67].

A geometry-aware alternative is the p -Wasserstein distance,

$$W_p(P, Q) = \left(\inf_{\pi \in \Pi(P, Q)} \int \|x - y\|^p d\pi(x, y) \right)^{1/p} \quad (18)$$

where:

- $W_p(P, Q)$ is the p -Wasserstein distance between distributions P and Q ;
- P and Q are the real and synthetic distributions, as in Eq. 11;
- $\Pi(P, Q)$ is the set of all joint distributions (couplings) on the product space whose marginal distributions are P and Q ;
- π is one such coupling; $\inf$ denotes the infimum over all admissible couplings;
- $x - y$ is the ground distance on the sample space, taken as the Euclidean norm;
- $p \geq 1$ is the order of the distance; the special case $p = 1$ is the earth-mover distance.

For imaging applications, the Fréchet Inception Distance (and its omics analogue, the Fréchet Biological Distance) compares the means and covariances of Inception or encoder feature embeddings,

$$FID(P, Q) = \|\mu_P - \mu_Q\|^2 + Tr(\Sigma_P + \Sigma_Q - 2\sqrt{\Sigma_P \Sigma_Q}) \quad (19)$$

where:

- $FID(P, Q)$ is the Fréchet Inception (or Biological) Distance between distributions P and Q ;
- P and Q are the real and synthetic distributions
- μ_P and μ_Q are the mean feature vectors of real and synthetic samples in the Inception (or encoder) embedding space;
- $\mu_P - \mu_Q^2$ is the squared Euclidean distance between these means;
- Σ_P and Σ_Q are the corresponding feature-covariance matrices;

- $\text{Tr}(\cdot)$ is the matrix trace;
- $\sqrt{(\Sigma_P \Sigma_Q)}$ denotes the principal matrix square root of the product $\Sigma_P \Sigma_Q$;
- each feature distribution is modelled as multivariate Gaussian [68].

Downstream utility is quantified through Train-on-Synthetic, Test-on-Real (TSTR) experiments, where performance on real held-out data confirms that synthetic samples encode clinically relevant signal [69]. Biological fidelity requires preservation of cohort-level differentially expressed genes, pathway-enrichment concordance with the source cohort, and retention of established histological marker structure [70]. Fairness diagnostics subgroup distributional parity and bias-amplification ratios across age, sex, and ancestry are mandatory to prevent demographic skew magnification. Privacy audits encompassing membership-inference attacks and nearest-neighbour latent-space distance accompany every synthetic data release to preclude inadvertent re-identification [71]. Generative-augmentation studies in rare-cancer settings must publish the complete validation suite alongside model code as a condition of responsible reporting.

Dynamic models for optimization

Reinforcement learning (RL) is a paradigm of machine learning where the sequential decision-making process is modelled [30]. RL algorithms learn decision-making policies when they interact with the simulated environment or real data, balancing different objectives in the process, such as efficacy and toxicity [26]. In translational oncology research, the role of RL is in supporting adaptive trial design and optimizing therapy at the personal level through prescriptive capabilities that complement model predictions [34]. Along with real-world data feedback, RL enables continuous learning in the deployment phase of the Pipeline [37].

AI/ML applications in preclinical oncology and translational target discovery

Preclinical oncological modelling provides the fundamental translational substrate for establishing mechanistic biological validity and optimizing therapeutic interventions prior to clinical deployment. Conventional in vitro and in vivo workflows are constrained by high experimental variance, limited scalable throughput, and poor baseline predictive fidelity across species. Deep learning pipelines are increasingly deployed to map high-dimensional biochemical feature spaces, enhancing computational model realism and aligning laboratory discoveries with human physiological targets [36].

A critical translational bottleneck emerges at this interface: predictive networks optimized under controlled in silico or cell-line environments routinely exhibit

significant performance degradation when confronted with the stochastic, multi-layered biological complexity of real-world human malignancies. This generalization failure reflects a systematic mismatch between the statistical regularity of curated laboratory data and the phenotypic heterogeneity encountered in primary clinical tumor specimens [5, 29].

Multi-omics integration and computational biomarker discovery

High-throughput multi-omics discovery pipelines generate large, multi-layered data matrices spanning genomic variant arrays, transcriptomic count matrices, mass-spectrometry proteomics profiles, and epigenomic landscapes across heterogeneous tumour models. Graph neural networks and transformer-based encoders are increasingly preferred for reconstructing global signalling network topologies and isolating candidate oncogenic driver mutations, capturing relational dependencies between molecular entities that conventional unimodal analyses cannot represent. Unsupervised and semi-supervised embedding algorithms further map multi-parameter drug-sensitivity matrices and rank putative therapeutic targets according to biological relevance, druggability, and translational potential [72].

A fundamental limitation of these screening pipelines, however, is structural overfitting to uniform, immortalised cell lines maintained in artificially stable microenvironments. When an optimisation algorithm overfits its internal parameters to these controlled conditions, predictive calibration degrades substantially upon exposure to the complex cellular cross-talk, hypoxia gradients, and structural heterogeneity of primary human tumour tissue. Computational target ranking therefore cannot serve as a standalone translational endpoint; rigorous cross-platform validation arrays are required to establish authentic clinical utility [72].

Generative molecular design and chemical space navigation

Advanced machine learning has transformed early-stage drug design by shifting computational chemistry away from empirical trial-and-error workflows. Multi-layered analytical pipelines project high-dimensional chemical spaces to estimate binding affinity, pharmacokinetics, and off-target cross-reactivity profiles before chemical synthesis is initiated. Deep generative architectures including variational autoencoders, generative adversarial networks, and molecular diffusion models enable the de novo engineering of molecular structures optimised against multi-objective fitness functions encompassing target potency, selectivity, and minimal systemic toxicity [73].

The primary computational failure mode within these pipelines remains structural hallucination: generative algorithms trained on sparse or structurally homogeneous datasets frequently produce molecular graphs with apparently favourable binding metrics that violate foundational laws of physical chemistry, yielding compounds that are thermodynamically unstable or synthetically inaccessible. Mitigating this requires explicit integration of hard geometric, spatial, and quantum-mechanical constraints directly into latent-space optimisation layers, combined with independent synthetic-accessibility scoring and experimental wet-laboratory confirmation of computationally prioritised candidates [73].

Cost-sensitive machine learning for in silico toxicity and safety profiling

Unanticipated clinical toxicity remains the dominant driver of late-stage drug attrition, positioning early safety profiling as a critical economic and ethical necessity. Deep in silico toxicology networks process molecular graphs and legacy bioassay matrices to flag major adverse events including hepatotoxicity, cardiotoxicity, and genotoxicity during initial compound triage. By mapping structural chemical descriptors through deep neural networks, researchers execute high-throughput screening across large molecular libraries, prioritising viable candidates while reducing dependence on resource-intensive animal models [74].

This predictive paradigm is heavily constrained by severe data sparsity and systematic reporting bias: pharmaceutical developers rarely publish detailed profiles of failed, toxic structures, causing training data to remain disproportionately weighted toward successful, non-toxic molecules. Standard classifiers consequently miss niche off-target toxic signals due to this class imbalance. Reliable safety models require cost-sensitive learning frameworks applying asymmetric loss weights to heavily penalise false-negative toxic predictions, coupled with validation against structurally diverse, independently curated datasets. Rigorous selection of descriptive parameters and comprehensive validation across diverse structural datasets are paramount for ensuring predictive generalisability of computational safety frameworks [43, 74].

Cross-species transcriptomic alignment and translational domain Adaptation

To approximate human clinical physiology, preclinical workflows employ patient-derived organoids and multi-species animal systems as advanced translational platforms. Computer vision and machine learning enhance these models by automating high-content image-screening pipelines, eliminating observer bias from the tracking of cellular growth kinetics, morphometric changes,

and phenotypic drug responses. However, murine and human tumour microenvironments differ fundamentally in immune architecture, stromal composition, metabolic clearance rates, and vascular organization creating a profound cross-species transcriptomic divide that computational approaches must explicitly bridge [54, 75].

Bridging this divide requires transitioning from simple empirical models toward unsupervised domain adaptation and transfer learning protocols. These strategies enable a network to first learn core biological relationships within the animal model domain and then mathematically align latent features to human clinical endpoints, transforming isolated laboratory outputs into generalisable translational pipelines. Advanced transcriptomic profiling and robust gene expression validation are critical for accurately interpreting the complex genetic variations driving tumour behaviour across biological models a principle empirically demonstrated through rigorous cross-species transcriptomic analyses revealing fundamental regulatory divergences between murine and human tumour ecosystems [54]. Cross-species model alignment therefore represents an integrated computational and biological validation challenge, requiring convergent genomic, transcriptomic, and phenotypic evidence before preclinical findings can be extrapolated reliably to clinical trial design [32, 54, 75].

Global adoption of AI in oncology decision-making

One of the widely cited real-world examples of AI-based clinical decision support in oncology is IBM Watson for Oncology, developed in collaboration with Memorial Sloan Kettering Cancer Centre (USA). The system applies machine learning and natural language processing to analyze large volumes of structured and unstructured healthcare data, including electronic health records, genomic information, clinical guidelines, and published scientific literature. Watson for Oncology has been implemented in hospitals across multiple countries including the United States, China, South Korea, Thailand, and Australia, where it assists clinicians in selecting evidence-based and personalized treatment strategies. The system demonstrates how large-scale biomedical big data can be translated into clinically actionable insights, supporting oncologists in managing complex treatment decisions and improving consistence [76]. Subsequent independent evaluations, however, raised substantive concerns regarding the system's documented reliance on synthetic training scenarios, limited training-data transparency, and inconsistent concordance with local tumour board recommendations across international deployments; these limitations and broader lessons from high-profile clinical AI failures are critically examined in Sect. 7.2.

Clinical limitations and evidence gaps for IBM Watson for oncology

Several retrospective concordance studies evaluating IBM Watson for Oncology have reported considerable variability in agreement with clinician-recommended treatment decisions across cancer types and healthcare settings. These discrepancies were influenced by regional treatment practices, local clinical guidelines, drug accessibility, healthcare infrastructure, and variations in regulatory approval pathways across countries. Reliance on concordance alone as an indicator of clinical effectiveness has been increasingly questioned, as agreement with oncologist recommendations does not necessarily translate into improved patient outcomes or optimal therapeutic decision-making. Concerns regarding the system's limited transparency and opaque reasoning architecture further reduce clinician confidence and complicate real-world interpretation of treatment recommendations. Collectively, these limitations underscore the need for rigorous prospective validation, outcome-based assessment, and regional adaptation of AI decision-support systems prior to large-scale oncology deployment [77, 78].

Bridging the cross-species translational divide: computational strategies

A persistent contributor to the high attrition rate of oncology drug candidates exceeding 90% in late-stage development is the fundamental divergence between murine and human tumour biology, arising from orthology mismatches, species-specific paralog expansion, and substantially different immune and stromal microenvironments. The Found In Translation (FIT) framework uses paired mouse and human expression compendia to extrapolate murine differential-expression results to equivalent human conditions with measurable concordance gains [79]. Unsupervised domain-adaptation networks learn species-invariant latent representations through adversarial or optimal-transport objectives, while cross-species single-cell alignment tools such as SAMap resolve homologous cell states across biological models [80]. Advanced transcriptomic profiling and robust gene expression validation are critical for accurately interpreting the complex genetic variations driving tumour behaviour across different species [54, 81]. Integrated transfer-learning protocols that pre-train on large murine pharmacological screens and fine-tune on smaller human cohorts represent the most actively pursued path to reduced oncology drug attrition.

Gap: from algorithm development to clinical implementation

The translational gap in AI/ML-driven oncology refers to the persistent disparity between rapid advances in algorithm development and the comparatively limited adoption of these technologies in routine clinical practice [5, 26, 35]. Although substantial progress has been achieved in predictive model development and performance benchmarking, disproportionately less emphasis has been placed on model robustness, external generalisability, and real-world implementation readiness thereby limiting sustained clinical translation across diverse healthcare systems [26]. The global distribution and emerging trends in AI/ML applications in cancer research, illustrated in Fig. 4, reflect this systemic imbalance between publication output and operational clinical deployment.

Several challenges continue to obstruct practical implementation of AI/ML systems in oncology. Models are predominantly developed using retrospective, single-institution datasets that inadequately represent the demographic heterogeneity, workflow variability, and clinical complexity of routine practice [7]. Beyond dataset limitations, critical barriers include data quality deficits, interoperability failures between legacy electronic health record architectures and AI modules, fragmented health informatics infrastructure, and variable healthcare-system digital maturity each independently disrupting the path from validated algorithm to deployed clinical tool [8, 35].

Regulatory approval and governance present additional structural barriers. Existing frameworks were designed for static, locked software and are poorly equipped to address model drift, limited external validation, and continuous algorithm updates intrinsic to adaptive AI. In India, AI-enabled clinical software is regulated under the Medical Device Rules 2017 by CDSCO, requiring technical documentation and clinical performance evidence. The FDA's Predetermined Change Control Plan permits pre-authorised adaptive modifications under defined Algorithm Change Protocols, while the EU AI Act classifies oncology AI as High-Risk and mandates transparency, human oversight, and post-market surveillance. Harmonising these three regulatory regimes establishing standardised drift thresholds and audit requirements is an urgent governance priority [35, 77].

Although AI/ML technologies have demonstrated considerable potential across cancer diagnosis, prognosis, treatment planning, and precision oncology, real-world clinical impact is determined not by retrospective validation alone but by successful integration into routine workflows. Deployment requires institutional readiness, workforce digital literacy, sustainable infrastructure, reimbursement mechanisms, and regulatory clearance

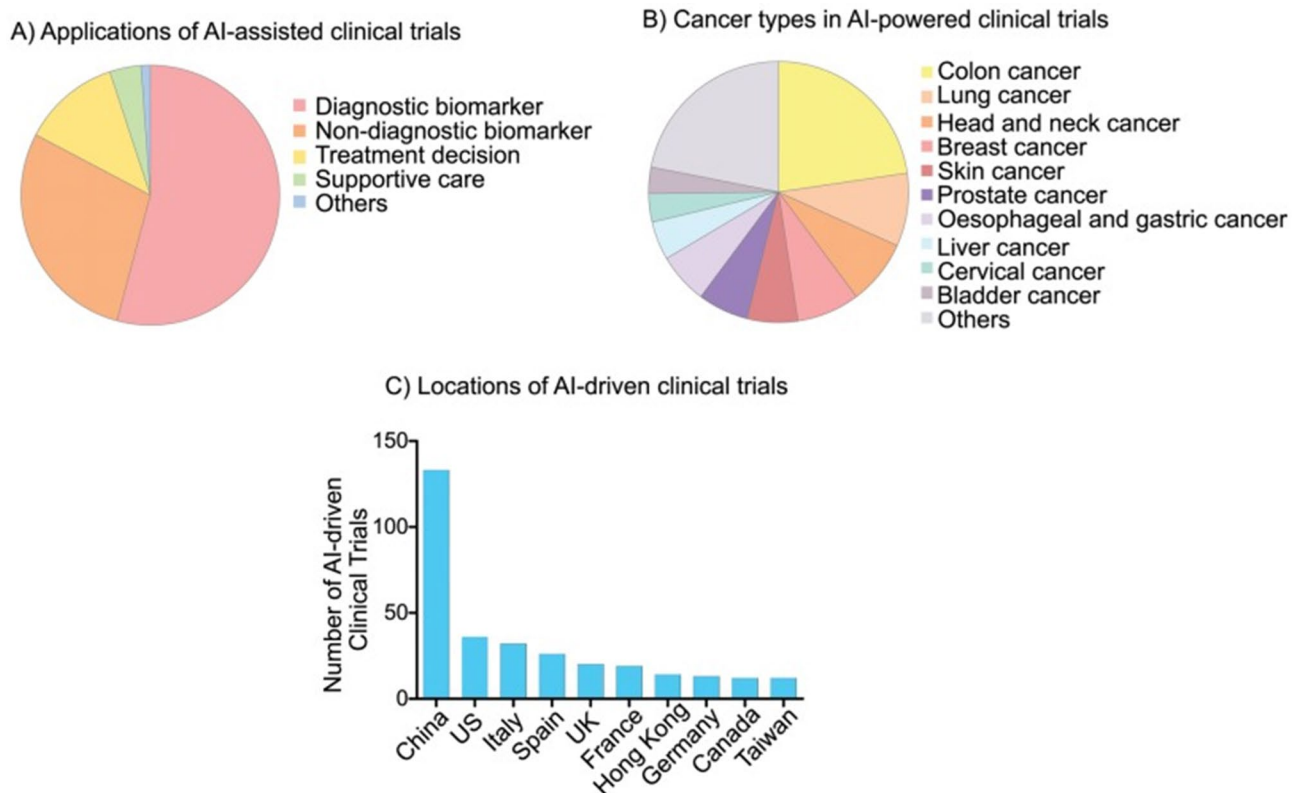


Fig. 4 Distribution of AI-assisted oncology clinical trials registered in ClinicalTrials.gov, stratified by application domain, cancer type, and geographic location. **(A)** Distribution across clinical domains: diagnostic biomarkers (~54%), non-diagnostic biomarkers (~28.6%), treatment decision-support (~12%), supportive care (~4%), and other applications. **(B)** Distribution by cancer type: colon cancer (~22.7%), lung cancer (~9%), head and neck cancer (~8%), breast cancer (~7.8%), skin cancer (~6.3%), prostate cancer (~6.3%), oesophageal and gastric cancer (~6.3%), liver cancer (~5%), cervical cancer (~3.5%), and bladder cancer (~3.3%). **(C)** Top ten countries by number of registered AI-assisted oncology trials evaluating AI-driven tools for cancer management. Reproduced from: Goda RY, Abdel-Aziz AK. Exploiting artificial intelligence in precision oncology: an updated comprehensive review. *J transl Med.* 2025;23:1397. doi:10.1186/s12967-025-07308-2, under the terms of the creative commons attribution 4.0 international license (CC BY 4.0)

a multidimensional challenge that publication-metric-focused research frameworks systematically fail to address. The gap between algorithmic performance and operational clinical utility will not close until implementation science is treated as a first-class research objective alongside model accuracy [35, 60].

Within India, disease burden, infrastructural heterogeneity, and variability in digital maturity across Tier-1, Tier-2, and rural facilities represent compounding implementation barriers. However, meaningful opportunities are emerging through context-aware deployment strategies and indigenous validation initiatives including Nirmai Thermaltyx for AI-assisted breast screening and the National Cancer Grid's federated biobanking infrastructure signalling a transition from theoretical AI adoption toward locally validated, population-adapted implementation [38].

Real-world clinical impact and implementation prerequisites

Although AI/ML technologies demonstrate strong performance in cancer diagnosis, prognosis, treatment planning, and precision oncology, real-world clinical impact depends on multiple factors beyond predictive accuracy. Clinician trust, workflow compatibility, healthcare infrastructure, cost-effectiveness, equitable access, and prospective multicenter validation jointly determine the practical utility and long-term sustainability of AI-assisted oncology systems. Future translational efforts must therefore prioritise not only performance improvement but also the development of clinically meaningful, safe, and equitable decision-support tools validated against real-world patient population heterogeneity across diverse healthcare settings [38, 60–62].

Integration of AI with telemedicine and mobile health platforms represents an important access-expansion strategy for geographically underserved populations [63, 82]. However, the core bottleneck in oncology AI research is a disproportionate emphasis on static retrospective

validation scores at the expense of long-term operational feasibility. Most published models train on highly curated historical repositories that mask the operational realities of active clinics. During prospective deployment, predictive reliability routinely degrades driven by unmitigated dataset shift, where changing patient demographics or new imaging hardware alter incoming feature distributions, and by algorithmic friction from poor integration with legacy electronic health record architectures [83, 84].

Clinical translation further fails because standard retrospective validation ignores human-system interaction dynamics. A model achieving exceptional retrospective AUC can still fail clinically when its interface increases cognitive burden, generating user fatigue or complete tool abandonment. Sustainable deployment therefore requires treating AI not as a software release but as a continuously monitored clinical process anchored by open-source auditing pipelines, standardised data schemas, prospective clinical utility testing, and explicit human-in-the-loop oversight. Marginal gains in raw predictive accuracy must be deprioritised in favour of operationally robust, clinician-compatible, governance-compliant translational systems [83, 84].

AI failures in oncology: documented lessons for robust translation

A scientifically balanced evaluation of oncology AI requires explicit engagement with documented implementation failures. IBM Watson for Oncology trained substantially on synthetic scenarios at a single institution demonstrated low concordance with tumour board recommendations across multiple international deployments, ultimately contributing to its commercial divestment [85]. The Epic Sepsis Model, deployed across hundreds of US hospitals, achieved a sensitivity of only 33% and AUC of 0.63 in independent external validation far below developer-reported performance while generating clinically significant alert fatigue [86]. A systematic review of COVID-19 imaging AI found none of 415 evaluated models clinically deployable, with data leakage, site-confounded labels, and demographic bias as recurrent failure mechanisms [33]. Dataset shift the distributional change between training and deployment populations has been identified as a generalised threat across clinical AI pipelines [59]. These failures collectively define minimum operational standards: prospective external validation on demographically and geographically distinct cohorts, site-stratified rather than patient-stratified data splits, explicit auditing for scanner, staining, site, and label-source confounders, continuous drift monitoring, and shadow-mode deployment prior to clinical workflow activation.

To systematically evaluate the clinical footprint of artificial intelligence within the oncology landscape, the dataset visualised in Fig. 2 was generated through a structured scoping audit of ClinicalTrials.gov. An initial registry query targeting machine learning, deep learning, and computational oncology applications was executed across all active trial records available at the time of audit. A subsequent data-cleansing protocol eliminated records lacking verified algorithmic intervention pipelines, duplicate entries, and trials without oncology-specific primary clinical endpoints. This filtering procedure yielded a finalised analytical cohort of 397 active, independently verified oncology clinical trials, constituting the quantitative basis for all reported distributions.

The reported percentage distributions were derived by stratifying the 397-trial cohort into mutually exclusive subsets according to primary clinical intent, cancer type, and geographic registration. For a finite cohort of N_{total} validated trials, let i denote a specific clinical application domain and n_i denote the number of trials classified within that domain. The normalised percentage distribution P_i for each category is formally defined as:

$$P_i = \left(\frac{n_i}{N_{\text{total}}} \right) \times 100 \quad (20)$$

and $N_{\text{total}} = 397$ and $\sum n_i = N_{\text{total}}$ where:

- P_i is the final percentage allocation representing the relative clinical footprint of the i -th application domain;
- n_i represents the absolute frequency count of active, verified clinical trials strictly categorized within that specific operational layer;
- N_{total} is the total number of unique, non-redundant clinical trials qualifying after data-cleansing filters ($N_{\text{total}} = 397$).

To confirm the structural composition of the dataset, the categorical sum must satisfy the constraint of completeness, ensuring no overlapping profiles or unclassified data remnants exist within the analyzed ecosystem:

$$\sum_{i=1}^M n_i = n_1 + n_2 + \dots + n_M = N_{\text{total}} \quad (21)$$

where M represents the total number of distinct clinical application categories ($M = 4$).

Substituting the empirically derived frequency values:

$$n_{\text{diagnostic}} = 214$$

$$n_{\text{non-diagnostic}} = 114$$

$$n_{\text{treatment}} = 48$$

$$n_{\text{supportive}} = 16$$

into the system demonstrates precise compliance with the total sample space boundaries, providing a verifiable, reproducible statistical foundation for the observed macro-trends. Diagnostic workflows comprised the clear majority at 54%, driven by a heavy focus on algorithmic lesion characterization and automated screening, followed by non-diagnostic monitoring protocols at 28.6%, automated treatment decision-support systems at 12%, and patient supportive care platforms at 4%.

To confirm the structural completeness of the audited dataset, the categorical sum across all application domains must satisfy $\sum n_i = N_{\text{total}} = 397$, ensuring no overlapping trial classifications or unclassified records persist within the analytical cohort. Compliance with this total-sample-space boundary condition provides a verifiable and reproducible statistical foundation for all reported distributional patterns. Diagnostic workflows comprised the clear majority at 54%, driven by algorithmic lesion characterization and automated screening pipelines; non-diagnostic monitoring protocols followed at 28.6%, automated treatment decision-support systems at 12%, and patient supportive care platforms at 4%, with the combined distribution confirming full categorical coverage.

India-specific implementation: translational challenges and contextual opportunities

To cover up the gap between research and clinical implementation, a well-thought-out implementation strategy should be developed that fits well with the Indian healthcare scenario.

Firstly, data standardization and interoperability should be enhanced with the help of FHIR and a unified national health data architecture to enable smooth integration between different healthcare centres [87, 88]. Secondly, capacity building and collaboration should also take centre stage with the help of AI literacy among clinicians and collaboration between data scientists and healthcare practitioners to ensure that the developed solutions are clinically relevant [89]. Thirdly, the regulatory framework should also change to accommodate AI-based medical technologies with the help of thorough validation using Indian population data for ensuring fairness, reliability, and generalizability [84, 90]. Lastly, public-private partnerships should also play a key role in accelerating innovation with the help of shared resources for the large-scale implementation of AI systems in the domain of oncology [91, 92].

The trajectory of AI/ML adoption in India will be shaped by scalability imperatives, cost-effectiveness constraints, and population health literacy factors that must collectively address the heterogeneous infrastructural landscape and fragmented data ecosystems characteristic of Indian healthcare. Expansion of national digital health initiatives and cancer registry networks is projected to enable construction of large, representative, and longitudinally annotated datasets, supporting more robust and externally generalisable AI/ML models [93, 94]. Federated and privacy-preserving learning frameworks represent a particularly critical research direction, enabling collaborative multi-institutional model development without centralised patient data transfer a framework directly suited to contexts where data sharing is frequently constrained by infrastructural and regulatory barriers [95, 96]. Future AI deployments must additionally prioritise lightweight, interpretable, and workflow-compatible architectures for early detection, triage, and treatment in rural and semi-urban settings, while maintaining regulatory alignment with CDSCO Medical Device Rules 2017 requirements for prospective clinical validation using representative Indian population data [61, 62, 97].

The vast majority of Western-trained oncology models lack adequate South Asian demographic representation, producing documented performance degradation when deployed in India due to distinct epidemiological realities including a high burden of region-specific oral squamous cell carcinoma subtypes and early-onset breast cancers presenting in dense fibroglandular tissue. To address this translational gap, Indian initiatives are transitioning toward localised, prospectively validated deployment frameworks. Niramai Thermalytix has circumvented the known limitations of mammography in dense breast tissue by deploying AI-driven infrared thermography to detect metabolic heat variations, validated across multi-centre Indian trials [AW]. Concurrently, the contactless monitoring platform Dozee employs machine learning to continuously track patient vitals, directly mitigating acute nurse workforce shortages at major centres including NIMHANS [AX]. Structural data fragmentation is being systematically countered by the National Cancer Grid, linking over 300 hospitals under unified data schemas to build India-native biobanks essential for training population-representative oncology AI [98, 99]. These initiatives collectively represent a transition from Western-adapted deployment toward indigenously validated, epidemiologically grounded precision oncology frameworks.

Regulatory landscape for AI-driven healthcare technologies in India

In India, AI-based clinical software is regulated under the Medical Device Rules (MDR), 2017 by the Central

Table 2 Comparative regulatory frameworks governing adaptive artificial intelligence in oncology across the United States (FDA), European Union (EU), and India (CDSCO)

Dimension	United States (FDA)	European Union (EU)	India (CDSCO)	References
Primary instrument	FD&C Act, section 515C; PCCP Final Guidance, December 2024	MDR (EU) 2017/745; IVDR (EU) 2017/746; AI Act (EU) 2024/1689	Medical Device Rules, 2017 under Drugs and Cosmetics Act, 1940	[104–106]
Classification of clinical AI	Software as a Medical Device (SaMD), risk-classified Class I–III	High-risk AI system under AI Act, layered on MDR/IVDR Class IIa–III conformity	Medical device, Class A–D based on intended risk profile	[104–106]
Approval pathway	510(k), De Novo, or PMA depending on risk class; PCCP authorisation available for pre-specified algorithm changes	MDR/IVDR conformity assessment by Notified Body, plus separate AI Act conformity assessment	Manufacturing or import licence (Form MD-9/MD-15) following technical-file review by CDSCO	[104–106]
Adaptive-algorithm pathway	PCCP with SaMD Pre-Specification and Algorithm Change Protocol; pre-authorised modifications do not require new submission	AI Act Article 43 requires reassessment for substantial modifications; performance drift is an explicit risk-management obligation	No formal PCCP-equivalent; algorithms either locked at approval or resubmitted in full for each material update	[104–106]
Post-market obligations	Total Product Life Cycle (TPLC) approach: real-world performance monitoring, periodic reporting, and MDR adverse-event reporting	Mandatory post-market surveillance plan, periodic safety update reports, and serious-incident reporting under MDR/MDCG guidance and AI Act post-market monitoring	Post-market surveillance recommended but not yet mandated for AI; adverse-event reporting via Materialvigilance Programme of India (MvPI)	[104–106]
Transparency and data governance	Algorithm-modification transparency required in labelling and Special Controls; bias-mitigation principles incorporated in 2024 PCCP guidance	AI Act Articles 10–15: data governance, technical documentation, transparency, human oversight, robustness, and cybersecurity requirements	Technical documentation must include dataset description, validation evidence, algorithm design, and QMS practices; no codified AI-specific data-governance or bias-mitigation regime yet	[104–106]
Maturity for adaptive oncology AI	Most mature framework for adaptive AI; PCCP specifically designed for continuously evolving models	Comprehensive but newest framework; many AI Act provisions phase in progressively through 2026–2027	Foundational medical-device framework in place; AI-specific and adaptive-algorithm governance provisions are still emerging	[104–106]

Drugs Standard Control Organization (CDSCO). When artificial intelligence is intended for diagnostic or therapeutic decision-making, it is considered a medical device and requires submission of detailed technical documentation prior to approval. Manufacturers must specify the intended clinical use, risk classification, algorithm design, and evidence of analytical and clinical performance [100, 101]. Documentation related to dataset quality, safety validation, and quality management practices is also required to ensure reliability of AI-based outputs. Post-market monitoring is additionally recommended to evaluate real-world performance and maintain patient safety [102]. These regulatory provisions aim to ensure that AI-driven healthcare technologies are safe, effective, and suitable for clinical implementation in the Indian healthcare system [103].

The three principal regulatory ecosystems treat adaptive AI differently (Table 2). In the United States, the FDA’s December 2024 final guidance on Predetermined Change Control Plans under section 515C of the FD&C Act permits manufacturers of AI/ML-enabled SaMD to obtain prospective authorisation for pre-specified modifications through a SaMD Pre-Specification and an Algorithm Change Protocol [104]. In the European Union, Regulation (EU) 2024/1689 (the AI Act) classifies most medical AI as high-risk, imposing obligations

on data governance, transparency, human oversight, and post-market monitoring above existing MDR or IVDR conformity requirements; adaptive learning systems additionally require explicit risk management provisions for performance drift and distributional shift [105]. India’s Medical Device Rules 2017, administered by CDSCO, regulate AI-based clinical software through risk-classified pathways but lack a formal PCCP-equivalent mechanism adaptive algorithms must therefore either remain locked at approval or undergo full repeat submissions for material updates [106]. Harmonizing the Indian framework with internationally accepted adaptive-algorithm governance principles and post-market surveillance standards would accelerate domestic oncology AI translation while preserving patient safety. A structured comparison across dimensions most relevant adaptive AI is provided in Table 2.

Conclusion

Artificial intelligence and machine learning have substantial potential to improve translational oncology, but their value lies less in isolated predictive performance than in their integration across the full pathway from data generation to real-world clinical use. The central message of this review is that successful oncology AI requires coordinated attention to multimodal data quality,

harmonization, biologically credible modelling, rigorous and multidimensional validation, workflow integration, and adaptive governance.

By framing the manuscript as a narrative review and end-to-end translational framework article, this revision clarifies both scope and distinctive contribution. It highlights that the true bottleneck in oncology AI is not a shortage of algorithms, but a shortage of clinically grounded, externally validated, implementation-ready systems. Addressing that bottleneck will require interoperable data ecosystems, federated collaboration, explainable and auditable models, and prospective evidence of patient benefit. For India and other heterogeneous healthcare settings, context-sensitive deployment strategies will be essential if AI/ML is to move from computational promise to measurable oncology impact [107, 108].

Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
ANN	Artificial Neural Network
DNN	Deep Neural Network
CNN	Convolutional Neural Network
GNN	Graph Neural Network
RL	Reinforcement Learning
NLP	Natural Language Processing
LLM	Large Language Model
GAN	Generative Adversarial Network
DANN	Domain-Adversarial Neural Network
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
AUROC	Area Under the Receiver Operating Characteristic Curve
AUPRC	Area Under the Precision–Recall Curve
PR	Precision–Recall
MCC	Matthews Correlation Coefficient
ECE	Expected Calibration Error
TPR	True Positive Rate
FPR	False Positive Rate
DCA	Decision Curve Analysis
NB	Net Benefit
LASSO	Least Absolute Shrinkage and Selection Operator
EN	Elastic Net
ComBat	Empirical Bayes Batch-Effect Correction Method
MMD	Maximum Mean Discrepancy
FID	Fréchet Inception Distance
MOFA	Multi-Omics Factor Analysis
TSTR	Train-on-Synthetic, Test-on-Real
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations
PRS	Polygenic Risk Score
RT-DMF	Residual Threshold Deep Matrix Factorization
DIPK	Deep Neural Network Integrating Prior Knowledge
HER	Electronic Health Record
EMR	Electronic Medical Record
FHIR	Fast Healthcare Interoperability Resources
API	Application Programming Interface
PACS	Picture Archiving and Communication System
GDPR	General Data Protection Regulation
DNA	Deoxyribonucleic Acid
RNA	Ribonucleic Acid
mRNA	Messenger Ribonucleic Acid
NGS	Next-Generation Sequencing
WES	Whole-Exome Sequencing
WGS	Whole-Genome Sequencing
SNP	Single-Nucleotide Polymorphism

SNV	Single-Nucleotide Variant
CNV	Copy-Number Variation
TCGA	The Cancer Genome Atlas
ICGC	International Cancer Genome Consortium
GDSC	Genomics of Drug Sensitivity in Cancer
CCLL	Cancer Cell Line Encyclopedia
MRI	Magnetic Resonance Imaging
CT	Computed Tomography
PET	Positron Emission Tomography
WSI	Whole-Slide Image
H&E	Haematoxylin and Eosin
FDA	Food and Drug Administration
EU	European Union
CDSCO	Central Drugs Standard Control Organisation
MDR	Medical Device Rules
PCCP	Predetermined Change Control Plan
TPLC	Total Product Life Cycle
SaMD	Software as a Medical Device
NHS	National Health Service
WHO	World Health Organization
IBM WFO	IBM Watson for Oncology
MSKCC	Memorial Sloan Kettering Cancer Center
NCCN	National Comprehensive Cancer Network
FIT	Found In Translation
CONSORT-AI	Consolidated Standards of Reporting Trials–Artificial Intelligence
DECIDE-AI	Developmental and Exploratory Clinical Investigations of Decision Support Systems Driven by Artificial Intelligence

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Author contributions

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References

- Jiang P, Sinha S, Aldape K, Hannenhalli S, Sahinalp C, Rupp R. Big data in basic and translational cancer research. *Nat Rev Cancer*. 2022;22(11):625–39. <https://doi.org/10.1038/s41568-022-00502-0>.
- Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clinicians*. 2024;74(3):229–63. <https://doi.org/10.3322/caac.21834>.
- Zhu L, Zhu J, Wang Q, Sun X, Yuan Y, Ding S, et al. Global, regional, and national burden of 34 cancer groups across 204 countries and territories, 1990–2021, and projections to 2050: a systematic analysis of the Global burden of disease study 2021. *Front Oncol*. 2025;15:1660125. <https://doi.org/10.3389/fonc.2025.1660125>.
- Costa FF. Big data analytics in oncology: opportunities and challenges. *Meth-ods*. 2016;111:50–56.
- Bhinder B, Gilvary C, Madhukar NS, Elemento O. Artificial intelligence in cancer research and precision medicine. *Cancer Discov*. 2021;11(4):900–15. <https://doi.org/10.1158/2159-8290.CD-21-0090>.
- Kahn CE Jr, Carrino JA, Flynn MJ, Peck DJ, Horii SC. DICOM and radiology: past, present, and future. *J Am Coll Radiol*. 2007;4(9):652–57. <https://doi.org/10.1016/j.jacr.2007.06.004>.
- Gogoshin G, Rodin AS. Graph neural networks in cancer and oncology research: emerging and future trends. *Cancers (Basel)*. 2023;15(24):5858. <https://doi.org/10.3390/cancers15245858>.
- Davenport T, Kalakota R. The potential for artificial intelligence in healthcare. *Future Healthcare J*. 2019;6(2):94–98. <https://doi.org/10.7861/futurehosp.6-2-94>.
- Chaudhary K, Poirion OB, Lu L, Garmire LX. Deep learning-based multi-omics integration robustly predicts survival in liver cancer. *Clin Cancer Res*. 2018;24(6):1248–59. <https://doi.org/10.1158/1078-0432.CCR-17-0853>.
- Masucci M, Karlsson C, Blomqvist L, Ernberg I. Bridging the divide: a review on the implementation of personalized cancer medicine. *JPM*. 2024;14(6):561. <https://doi.org/10.3390/jpm14060561>.
- Sweeney SM, Hamadeh HK, Abrams N, Adam SJ, Brenner S, Connors DE, et al. Case studies for overcoming challenges in using big data in cancer. *Cancer Res*. 2023;83(8):1183–90. <https://doi.org/10.1158/0008-5472.CAN-22-1277>.
- Frasca M, La Torre D, Repetto M, De Nicolo V, Pravattoni G, Cutica I. Artificial intelligence applications to genomic data in cancer research: a review of recent trends and emerging areas. *Discov Anal*. 2024;2(1):10. <https://doi.org/10.1007/s44257-024-00017-y>.
- Stokes JM, Yang K, Swanson K, Jin W, Cubillos-Ruiz A, Donghia NM, et al. A deep learning approach to antibiotic discovery. *Cell*. 2020;181(2):475–83. <https://doi.org/10.1016/j.cell.2020.04.001>.
- Boehm KM, Khosravi P, Vanguri R, Gao J, Shah SP. Harnessing multimodal data integration to advance precision oncology. *Nat Rev Cancer*. 2022;22(2):114–26. <https://doi.org/10.1038/s41568-021-00408-3>.
- Witten DM, Tibshirani R. Survival analysis with high-dimensional covariates. *Stat Methods Med Res*. 2010;19(1):29–51. <https://doi.org/10.1177/0962280209105024>.
- Simon N, Friedman JH, Hastie T, Tibshirani R. Regularization paths for Cox's proportional hazards model via coordinate descent. *J Stat Soft*. 2011;39(5):1–13. <https://doi.org/10.18637/jss.v039.i05>.
- Wang P, Li Y, Reddy CK. Machine learning for survival analysis: a survey. *ACM Comput Surv*. 2019;51(6):1–36. <https://doi.org/10.1145/3214306>.
- Liu F, Beck S, Yang L, Luo H, Zhang K. Advancing AI for multi-omics and clinical data integration in basic and translational cancer research. *Nat Rev Cancer*. 2026;26(5):285–301. <https://doi.org/10.1038/s41568-026-00922-2>.
- Leek JT, Scharpf RB, Bravo HC, Simcha D, Langmead B, Johnson WE, et al. Tackling the widespread and critical impact of batch effects in high-throughput data. *Nat Rev Genet*. 2010;11(10):733–39. <https://doi.org/10.1038/nrg2825>.
- Picard M, Scott-Boyer MP, Bodein A, Perin O, Droit A. Integration strategies of multi-omics data for machine learning analysis. *Comput Struct Biotechnol J*. 2021;19:3735–46. <https://doi.org/10.1016/j.csbj.2021.06.030>.
- Argelaguet R, Velten B, Arnol D, Dietrich S, Zenz T, Marioni JC, et al. Multi-omics factor analysis—a framework for unsupervised integration of multi-omics data sets. *Mol Syst Biol*. 2018;14(6):e8124. <https://doi.org/10.15252/msb.20178124>.
- Argelaguet R, Arnol D, Bredikhin D, Deloro Y, Velten B, Marioni JC, et al. MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data. *Genome Biol*. 2020;21(1):111. <https://doi.org/10.1186/s13059-020-02015-1>.
- Stahlschmidt SR, Ulfenborg B, Synnergren J. Multimodal deep learning for biomedical data fusion: a review. *Briefings Bioinf*. 2022;23(2):bbab569. <https://doi.org/10.1093/bib/bbab569>.
- Singhal K, Azizi S, Tu T, Mahdavi SS, Wei J, Chung HW, et al. Large language models encode clinical knowledge. *Nature*. 2023;620(7972):172–80. <https://doi.org/10.1038/s41586-023-06291-2>.
- Vamathevan J, Clark D, Czodrowski P, Dunham I, Ferran E, Lee G, et al. Applications of machine learning in drug discovery and development. *Nat Rev Drug Discov*. 2019;18(6):463–77. <https://doi.org/10.1038/s41573-019-0024-5>.
- Tjoa E, Guan C. A survey on explainable artificial intelligence (XAI): toward medical XAI. *IEEE Trans Neural Netw Learn Syst*. 2021;32(11):4793–813. <https://doi.org/10.1109/TNNLS.2020.3027314>.
- Kourou K, Exarchos TP, Exarchos KP, Karamouzis MV, Fotiadis DI. Machine learning applications in cancer prognosis and prediction. *Comput Struct Biotechnol J*. 2015;13:8–17. <https://doi.org/10.1016/j.csbj.2014.11.005>.
- Litjens G, Kooi T, Bejnordi BE, Setio AAA, Ciampi F, Ghafoorian M, et al. A survey on deep learning in medical image analysis. *Med Image Anal*. 2017;42:60–88. <https://doi.org/10.1016/j.media.2017.07.005>.
- Gao H, Korn JM, Ferretti S, Monahan JE, Wang Y, Singh M, et al. High-throughput screening using patient-derived tumor xenografts to predict clinical trial drug response. *Nat Med*. 2015;21(11):1318–25. <https://doi.org/10.1038/nm.3954>.
- Li T, Sahu AK, Talwalkar A, Smith V. Federated learning: challenges, methods, and future directions. *IEEE Signal Process Mag*. 2020;37(3):50–60. <https://doi.org/10.1109/MSP.2020.2975749>.
- Goel I, Bhaskar Y, Kumar N, Singh S, Amanullah M, Dhar R, et al. Role of AI in empowering and redefining the oncology care landscape: perspective from a developing nation. *Front Digit Health*. 2025;7:1550407. <https://doi.org/10.3389/fdgth.2025.1550407>.
- Rajpurkar P, Chen E, Banerjee O, Topol EJ. AI in health and medicine. *Nat Med*. 2022;28(1):31–38. <https://doi.org/10.1038/s41591-021-01614-0>.
- Roberts M, Driggs D, Thorpe M, Gilbey J, Yeung M, Ursprung S, et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nat Mach Intell*. 2021;3(3):199–217. <https://doi.org/10.1038/s42256-021-00307-0>.
- Liu R, Rizzo S, Whipple S, Pal N, Pineda AL, Lu M, et al. Evaluating eligibility criteria of oncology trials using real-world data and AI. *Nature*. 2021;592(7855):629–33. <https://doi.org/10.1038/s41586-021-03430-5>.
- Kelly CJ, Karthikesalingam A, Suleyman M, Corrado G, King D. Key challenges for delivering clinical impact with artificial intelligence. *BMC Med*. 2019;17(1):195. <https://doi.org/10.1186/s12916-019-1426-2>.
- Day CP, Merlino G, Van Dyke T. Preclinical mouse cancer models: a maze of opportunities and challenges. *Cell*. 2015;163(1):39–53. <https://doi.org/10.1016/j.cell.2015.08.068>.
- Sendak MP, D'Arcy J, Kashyap S, Gao M, Nichols M, Corey K, et al. A path for translation of machine learning products into healthcare delivery. *EMJ Innov*. 2020;4(1):45–53. <https://doi.org/10.33590/emjinnov/19-00172>.
- Naik N, Hameed BMZ, Shetty DK, Swain D, Shah M, Paul R, et al. Legal and ethical consideration in artificial intelligence in healthcare: who takes responsibility? *Front Surg*. 2022;9:862322. <https://doi.org/10.3389/fsurg.2022.862322>.
- Lipkova J, Chen RJ, Chen B, Lu MY, Barbieri M, Shao D, et al. Artificial intelligence for multimodal data integration in oncology. *Cancer Cell*. 2022;40(10):1095–110. <https://doi.org/10.1016/j.ccell.2022.09.012>.
- Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*. 2007;8(1):118–27. <https://doi.org/10.1093/biostatistics/kxj037>.
- Zhang Y, Parmigiani G, Johnson WE. ComBat-seq: batch effect adjustment for RNA-seq count data. *NAR Genomics Bioinf*. 2020;2(3):lqaa078. <https://doi.org/10.1093/nargab/lqaa078>.
- Guan H, Liu M. Domain adaptation for medical image analysis: a survey. *IEEE Trans Biomed Eng*. 2022;69(3):1173–85. <https://doi.org/10.1109/TBME.2021.3117407>.
- Schork NJ. Artificial intelligence and personalized medicine. *Cancer Treat Res*. 2019;178:265–83. https://doi.org/10.1007/978-3-030-16391-4_11.
- Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*. 2015;10(3):e0118432. <https://doi.org/10.1371/journal.pone.0118432>.
- Chicco D, Jurman G. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics*. 2020;21(1):6. <https://doi.org/10.1186/s12864-019-6413-7>.

46. Lin TY, Goyal P, Girshick R, He K, Dollar P. Focal loss for dense object detection. *IEEE Trans Pattern Anal Mach Intell*. 2020;42(2):318–27. <https://doi.org/10.1109/TPAMI.2018.2858826>.
47. Rana M, Ahmedi S, Fatima A, Ahmad S, Siddiqui N, Siddiqui N, et al. Synthesis, single crystal, TD-DFT, molecular dynamics simulation and DNA binding studies of carbothioamide analog. *J Mol Struct*. 2023;1287:135701. <https://doi.org/10.1016/j.molstruc.2023.135701>.
48. Vickers AJ, van Calster B, Steyerberg EW. A simple, step-by-step guide to interpreting decision curve analysis. *Diagn Progn Res*. 2019;3(1):18. <https://doi.org/10.1186/s41512-019-0064-7>.
49. Vickers AJ, Van Calster B, Steyerberg EW. Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests. *BMJ*. 2016;352:i6. <https://doi.org/10.1136/bmj.i6>.
50. Olawade DB, Oisakade EO, Bello OJ, Analikwu CC, Egbon E, Ojo A. Digital twins in oncology: from predictive modelling to personalised treatment strategies. *Crit Rev Oncol/Hematol*. 2026;220:105171. <https://doi.org/10.1016/j.critrevonc.2026.105171>.
51. Morley J, Machado CCV, Burr C, Cows J, Joshi I, Taddeo M, et al. The ethics of AI in healthcare: a mapping review. *Soc Sci Med*. 2020;260:113172. <https://doi.org/10.1016/j.socscimed.2020.113172>.
52. Singh V, Cheng S, Kwan AC, Ebinger J. United States Food and drug Administration regulation of clinical software in the era of artificial intelligence and machine learning. *Mayo Clinic Proc: Digit Health*. 2025;3(3):100231. <https://doi.org/10.1016/j.mcpdig.2025.100231>.
53. Sendak MP, Gao M, Brajer N, Balu S. Presenting machine learning model information to clinical end users with model facts labels. *NPJ Digit Med*. 2020;3(1):41. <https://doi.org/10.1038/s41746-020-0253-3>.
54. Koch LM, Baumgartner CF, Berens P. Distribution shift detection for the postmarket surveillance of medical AI algorithms: a retrospective simulation study. *NPJ Digit Med*. 2024;7(1):173. <https://doi.org/10.1038/s41746-024-01085-w>.
55. Abdul Razak MS, Nirmala CR, Sreenivasa BR, Lahza H, Lahza HFM. A survey on detecting healthcare concept drift in AI/ML models from a finance perspective. *J Healthc Inf Res*. 2025;9(4):312–35. <https://doi.org/10.1007/s41666-025-00482-w>.
56. Arun S, Grosheva M, Kosenko M, Robertus JL, Blyuss O, Gabe R, et al. Systematic scoping review of external validation studies of AI pathology models for lung cancer diagnosis. *NPJ Precis Onc*. 2025;9(1):100366. <https://doi.org/10.1038/s41698-025-00940-7>.
57. Seyyed-Kalantari L, Zhang H, McDermott MBA, Chen IY, Ghassemi M. Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. *Nat Med*. 2021;27(12):2176–82. <https://doi.org/10.1038/s41591-021-01595-0>.
58. Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*. 2019;366(6464):447–53. <https://doi.org/10.1126/science.aax2342>.
59. Finlayson SG, Subbaswamy A, Singh K, Bowers J, Kupke A, Zittrain J, et al. The clinician and dataset shift in artificial intelligence. *N Engl J Med*. 2021;385(3):283–86. <https://doi.org/10.1056/NEJMc2104626>.
60. de Vries CF, Lip G, Staff RT, Dymiter JA, Tse B, Ng A, et al. Prospective evaluation of artificial intelligence integration into breast cancer screening in multiple workflow settings: the GEMINI study. *Nat Commun*. 2024;15:6834. <https://doi.org/10.1038/s41467-024-51152-7>.
61. Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. *Nat Med*. 2019;25(1):44–56. <https://doi.org/10.1038/s41591-018-0300-7>.
62. Panch T, Szolovits P, Atun R. Artificial intelligence, machine learning and health systems. *J Global Health*. 2018;8(2):020303. <https://doi.org/10.7189/jogh.8.020303>.
63. World Health Organization. WHO guideline: recommendations on digital interventions for health system strengthening. 2019. Available from: <https://iris.who.int/server/api/core/bitstreams/c3c5f30-23cc-48d0-a3b5-c05ddd7f5349/content>. Geneva: World Health Organization.
64. Elmarakeby HA, Hwang J, Arafeh R, Crowdis J, Gang S, Liu D, et al. Biologically informed deep neural network for prostate cancer discovery. *Nature*. 2021;598(7880):348–52. <https://doi.org/10.1038/s41586-021-03922-4>.
65. Hartman E, Scott AM, Karlsson C, Mohanty T, Vaara ST, Linder A, et al. Interpreting biologically informed neural networks for enhanced proteomic biomarker discovery and pathway analysis. *Nat Commun*. 2023;14(1):5359. <https://doi.org/10.1038/s41467-023-41146-4>.
66. Famuyiwa SO, Ahmad S, Fakola EG, Olusola AJ, Adesida SA, Obagunle FO, et al. Comprehensive computational studies of naturally occurring kuguacins as antidiabetic agents by targeting visfatin. *Chem Afr*. 2023;6(3):1415–27. <https://doi.org/10.1007/s42250-023-00604-8>.
67. Grettton A, Borgwardt KM, Rasch MJ, Scholkopf B, Smola A. A kernel two-sample test. *J Mach Learn Res*. 2012;13(1):723–73.
68. Heusel M, Ramsauer H, Unterthiner T, Nessler B, Hochreiter S. Gans trained by a two time-scale update rule converge to a local Nash equilibrium. *Adv Neural Inf Process Syst*. 2017;30:6626–37. <https://doi.org/10.48550/arXiv.1706.08500>.
69. Esteban C, Hyland SL, Ratsch G. Real-valued (medical) time series generation with recurrent conditional GANs. *arXiv*: 1706.02633 [Preprint]. 2017. <https://doi.org/10.48550/arXiv.1706.02633>.
70. Treppner M, Salas-Bastos A, Hess M, Lenz S, Vogel T, Binder H. Synthetic single cell RNA sequencing data from small pilot studies using deep generative models. *Sci Rep*. 2021;11(1):9403. <https://doi.org/10.1038/s41598-021-88875-4>.
71. Giuffrè M, Shung DL. Harnessing the power of synthetic data in healthcare: innovation, application, and privacy. *NPJ Digit Med*. 2023;6(1):186. <https://doi.org/10.1038/s41746-023-00927-3>.
72. Buyse M, Sargent DJ, Grothey A, Matheson A, de Gramont A. Biomarkers and surrogate end points—the challenge of statistical validation. *Nat Rev Clin Oncol*. 2010;7(6):309–17. <https://doi.org/10.1038/nrclinonc.2010.43>.
73. Du Y, Jamasb AR, Guo J, Fu T, Harris C, Wang Y, et al. Machine learning-aided generative molecular design. *Nat Mach Intell*. 2024;6(6):589–604. <https://doi.org/10.1038/s42256-024-00843-5>.
74. Martinez-Outschoorn UE, Sotgia F, Lisanti MP. Caveolae and signalling in cancer. *Nat Rev Cancer*. 2015;15(4):225–37. <https://doi.org/10.1038/nrc3915>.
75. Sisk BA, Bereitschaft C, Enloe M, Schulz G, Mack J, DuBois J. Oncology clinicians' perspectives on online patient portal use in pediatric and adolescent cancer. *JCO Clin Cancer Inf*. 2023;7(7):e2300124. <https://doi.org/10.1200/CCI.23.00124>.
76. Somashekhar SP, Sepúlveda MJ, Puglielli S, Norden AD, Shortliffe EH, Rohit Kumar C, et al. IBM Watson for oncology and breast cancer treatment recommendations: agreement with an expert multidisciplinary tumor board. *Ann Oncol*. 2018;29(2):418–23. <https://doi.org/10.1093/annonc/mdx781>.
77. Tupasela A, Di Nucci E. Concordance as evidence in the Watson for oncology decision-support system. *AI Soc*. 2020;35(4):811–18. <https://doi.org/10.1007/s00146-020-00945-9>.
78. Zhou N, Zhang CT, Lv HY, Hao CX, Li TJ, Zhu JJ, et al. Concordance study between IBM Watson for oncology and clinical practice for patients with cancer in China. *Oncologist*. 2019;24(6):812–19. <https://doi.org/10.1634/theoncologist.2018-0255>.
79. Normand R, Du W, Briller M, Gaujoux R, Starosvetsky E, Ziv-Kenet A, et al. Found in translation: a machine learning model for mouse-to-human inference. *Nat Methods*. 2018;15(12):1067–73. <https://doi.org/10.1038/s41592-018-0214-9>.
80. Tarashansky AJ, Musser JM, Khariton M, Li P, Arendt D, Quake SR, et al. Mapping single-cell atlases throughout Metazoa unravels cell type evolution. *Elife*. 2021;10:e66747. <https://doi.org/10.7554/eLife.66747>.
81. Satyam R, Ahmad S, Raza K. Comparative genomic assessment of members of genus *tenacibaculum*: an exploratory study. *Mol Genet Genomics*. 2023;298(5):979–93. <https://doi.org/10.1007/s00438-023-02031-3>.
82. Scott Kruse C, Karem P, Shifflett K, Vegi L, Ravi K, Brooks M. Evaluating barriers to adopting telemedicine worldwide: a systematic review. *J Telemed Telecare*. 2018;24(1):4–12. <https://doi.org/10.1177/1357633X16674087>.
83. Yu Y, Mai Y, Zheng Y, Shi L. Assessing and mitigating batch effects in large-scale omics studies. *Genome Biol*. 2024;25(1):bbae567. <https://doi.org/10.1186/s13059-024-03401-9>.
84. Wiens J, Saria S, Sendak M, Ghassemi M, Liu VX VX, F, et al. Do no harm: a roadmap for responsible machine learning for healthcare. *Nat Med*. 2019;25(9):1337–40. <https://doi.org/10.1038/s41591-019-0548-6>.
85. Strickland E. IBM Watson, heal thyself: how IBM overpromised and underdelivered on AI healthcare. *IEEE Spectr*. 2019;56(4):24–31. <https://doi.org/10.1109/MSPEC.2019.8678513>.
86. Wong A, Otlés E, Donnelly JP, Krumm A, McCullough J, DeTroyer-Cooley O, et al. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Intern Med*. 2021;181(8):1065–70. <https://doi.org/10.1001/jamainternmed.2021.2626>.
87. Mandel JC, Kreda DA, Mandl KD, Kohane IS, Ramoni RB. SMART on FHIR: a standards-based, interoperable apps platform for electronic health records. *J Am Med Inf Assoc*. 2016;23(5):899–908. <https://doi.org/10.1093/jamia/ocv189>.

88. Lehne M, Sass J, Essenwanger A, Schepers J, Thun S. Why digital medicine depends on interoperability. *NPJ Digit Med*. 2019;2(1):79. <https://doi.org/10.1038/s41746-019-0158-1>.
89. Esteve A, Robicquet A, Ramsundar B, Kuleshov V, DePristo M, Chou K, et al. A guide to deep learning in healthcare. *Nat Med*. 2019;25(1):24–29. <https://doi.org/10.1038/s41591-018-0316-z>.
90. Benjamins S, Dhunoo P, Meskó B. The state of artificial intelligence-based FDA-approved medical devices and algorithms: an online database. *NPJ Digit Med*. 2020;3(1):118. <https://doi.org/10.1038/s41746-020-00324-0>.
91. Amann J, Blasimme A, Vayena E, Frey D, Madai VI. Explainability for artificial intelligence in healthcare: a multidisciplinary perspective. *BMC Med Inf Decis Mak*. 2020;20(1):310. <https://doi.org/10.1186/s12911-020-01332-6>.
92. Mallath MK, Taylor DG, Badwe RA, Rath GK, Shanta V, Pramesh CS, et al. The growing burden of cancer in India: epidemiology and social context. *The Lancet Oncol*. 2014;15(6):e205–12. [https://doi.org/10.1016/S1470-2045\(14\)70115-9](https://doi.org/10.1016/S1470-2045(14)70115-9).
93. Rajkumar A, Dean J, Kohane I. Machine learning in medicine. *N Engl J Med*. 2019;380(14):1347–58. <https://doi.org/10.1056/NEJMra1814259>.
94. Reddy S, Winter JS, Padmanabhan S. Artificial intelligence in healthcare: opportunities and challenges. *J Med Syst*. 2020;44(8):141. <https://doi.org/10.1007/s10916-020-01598-3>.
95. Yang Q, Liu Y, Chen T, Tong Y. Federated machine learning: concept and applications. *ACM Trans Intell Syst Technol*. 2019;10(2):1–19. <https://doi.org/10.1145/3298981>.
96. Rieke N, Hancox J, Li W, Milletari F, Roth HR, Albarqouni S, et al. The future of digital health with federated learning. *NPJ Digit Med*. 2020;3(1):119. <https://doi.org/10.1038/s41746-020-00323-1>.
97. Pati S, Baid U, Edwards B, Sheller M, Wang SH, Reina GA, et al. Federated learning enables big data for rare cancer boundary detection. *Nat Commun*. 2022;13(1):7346. <https://doi.org/10.1038/s41467-022-33407-5>.
98. Bansal R, Collison S, Krishnan L, Aggarwal B, Vidyasagar M, Kakileti ST, et al. A prospective evaluation of breast thermography enhanced by a novel machine learning technique for screening breast abnormalities in a general population of women presenting to a secondary care hospital. *Front Artif Intell*. 2023;5:512. <https://doi.org/10.3389/frai.2022.1050803>.
99. Kedia S, Rahman IU, Kaushal A, Mahizhvannan E, Jain RS, Radhakrishna R. Clinical validation of an indigenous micro-vibration vital parameter monitor Dozee VS. Proceedings of the IEEE European Biomedical Engineering Conference (EBEC). 2024. p. 114–19. <https://doi.org/10.1109/EBEC.2024.10423912>.
100. Dikshit R, Gupta PC, Ramasundarahettige C, Gajalakshmi V, Aleksandrowicz L, Badwe R, et al. Cancer mortality in India: a nationally representative survey. *Lancet*. 2012;379(9828):1807–16. [https://doi.org/10.1016/S0140-6736\(12\)60358-4](https://doi.org/10.1016/S0140-6736(12)60358-4).
101. Finlayson SG, Bowers JD, Ito J, Zittrain JL, Beam AL, Kohane IS. Adversarial attacks on medical machine learning. *Science*. 2019;363(6433):1287–89. <https://doi.org/10.1126/science.aaw4399>.
102. Bellamy RK, Dey K, Hind M, Hoffman SC, Houde S, Kannan K, et al. AI fairness 360: an extensible toolkit for detecting and mitigating algorithmic bias. *IBM J Res Dev*. 2019;63(4/5):4:1–4:15. <https://doi.org/10.1147/JRD.2019.2942287>.
103. Government of India. Ministry of Health and Family Welfare. Medical Devices Rules, 2017 under the Drugs and Cosmetics Act, 1940 [Internet]. New Delhi: Ministry of Health and Family Welfare; 2017 [cited 2026 Jun 22]. Available from: <https://cdsco.gov.in/>.
104. US Food and Drug Administration. Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions: final guidance for industry and FDA staff. 2024. Available from: <https://www.fda.gov/regulatory-information/search-fda-guidance-documents>. Silver Spring (MD): FDA.
105. European Parliament and Council. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act). *Off J Eur Union*. 2024;L1689:1–144.
106. Ministry of Health and Family Welfare, Government of India. Medical devices Rules. New Delhi: gazette of India, extraordinary, part II, section 3, sub-section (i); 31 January 2017. Available from: 2017. <https://cdsco.gov.in>.
107. Liu X, Cruz Rivera S, Moher D, Calvert MJ, Denniston AK, Ashrafian H, et al. Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension. *Lancet Digit Health*. 2020;2(10):e537–48. [https://doi.org/10.1016/S2589-7500\(20\)30218-1](https://doi.org/10.1016/S2589-7500(20)30218-1).
108. Vasey B, Nagendran M, Campbell B, Clifton DA, Collins GS, Denaxas S, et al. Reporting guideline for the early stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *BMJ*. 2022;377:e070904. <https://doi.org/10.1136/bmj-2022-070904>.

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